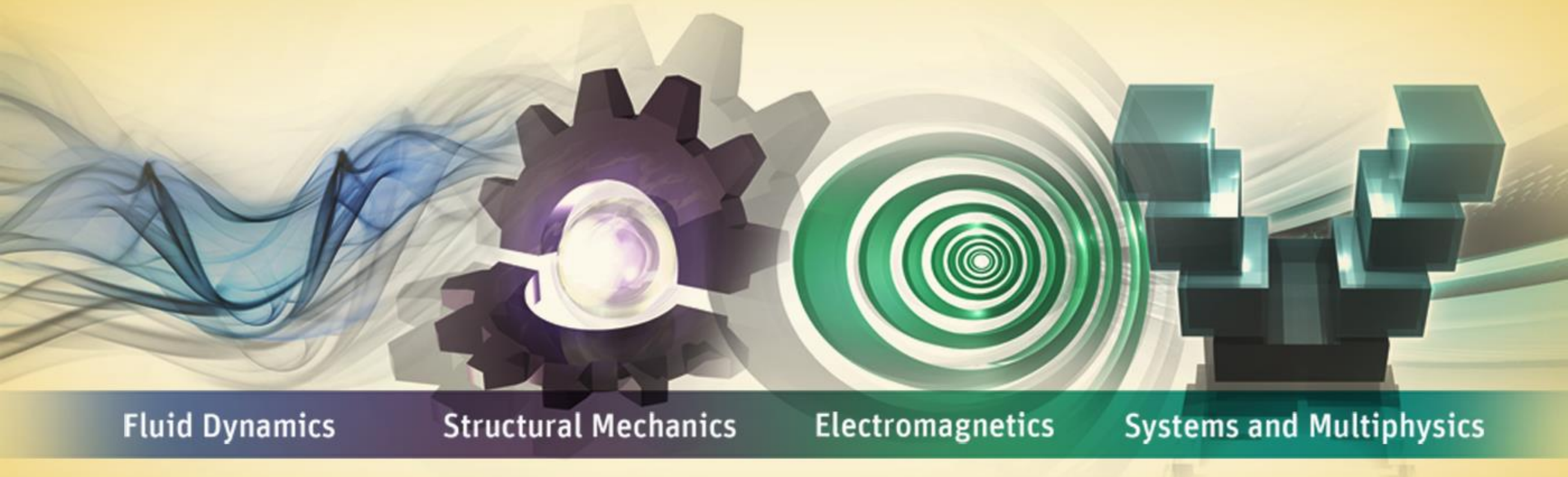
Realize Your Product Promise™

# ANSYS电源系统设计仿真平台




**ANSYS 中国**

# 内容提要

典型电源系统及其挑战

ANSYS电源系统设计平台实现

总结

# 电源系统应用领域

## 电源& 驱动器

电源变换/电网质量改善

驱动器

板上电源

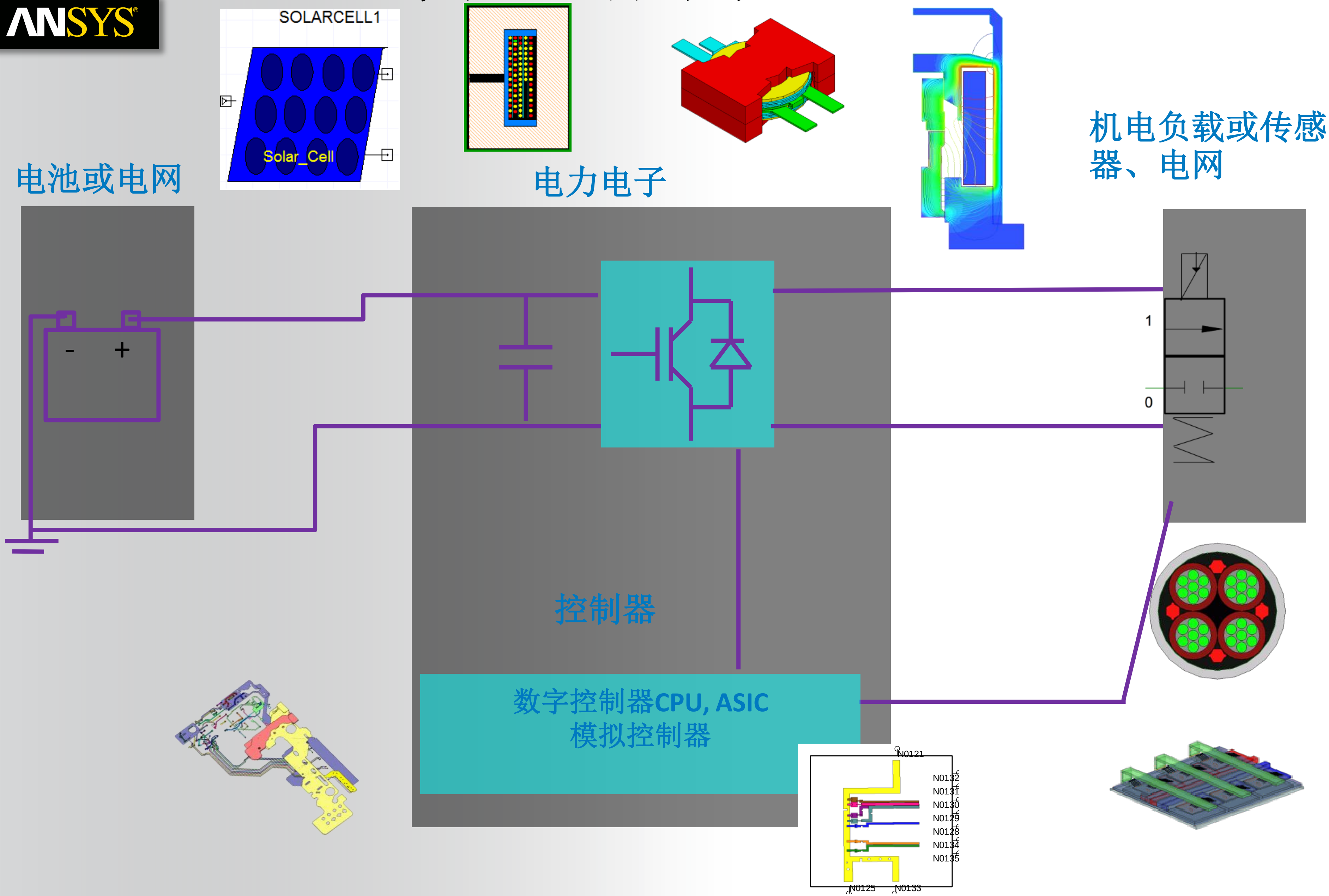

## 交通运输

航空航天

铁路运输

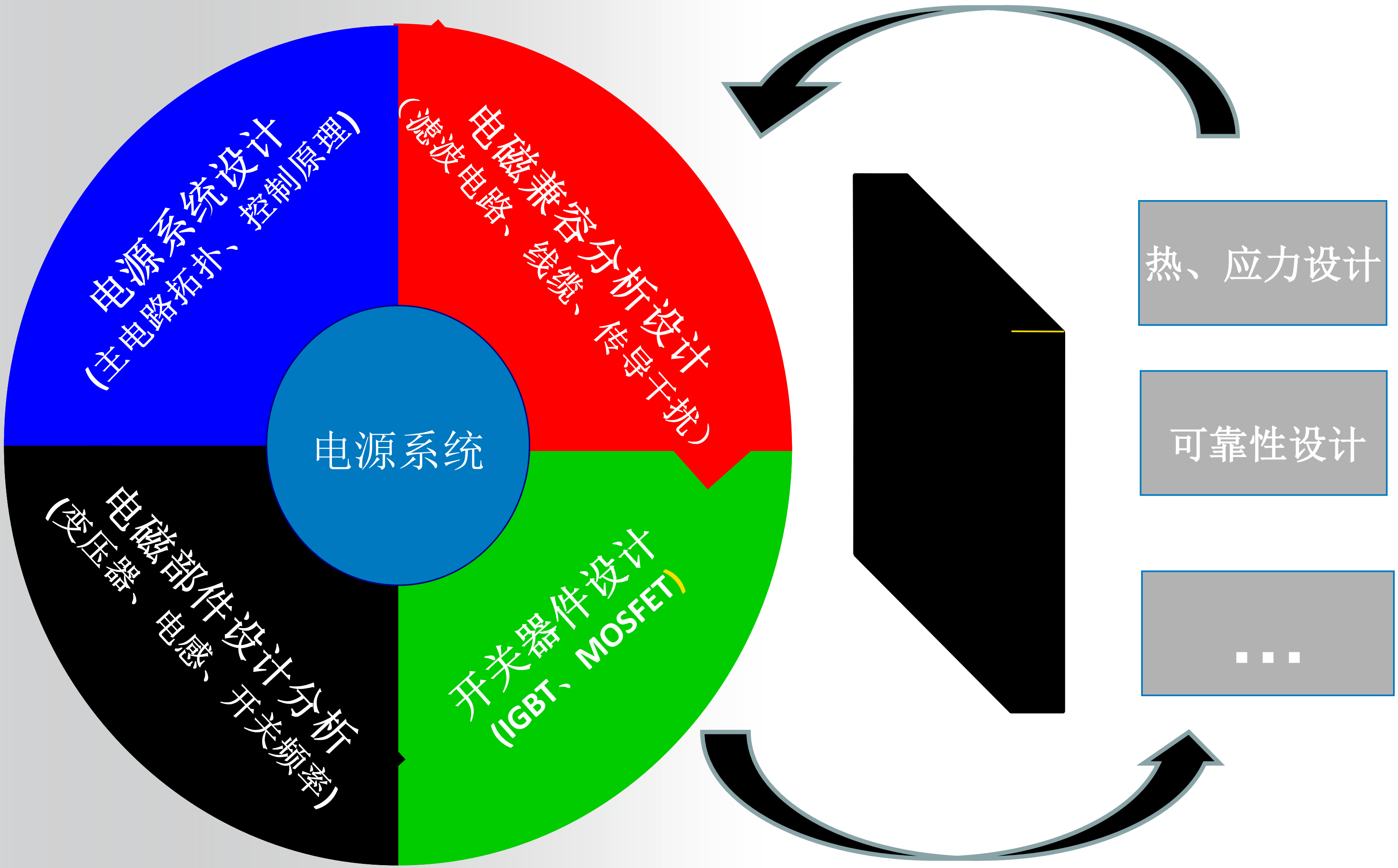
汽车

# 典型电源系统


SOLARCELL1
Solar_Cell
电池或电网
电力电子
机电负载或传感器、电网
-
+
1
0
控制器
数字控制器CPU, ASIC
模拟控制器
N0121
N0132
N0131
N0130
N0129
N0128
N0134
N0135
N0125
N0133

# 电源系统面临的挑战

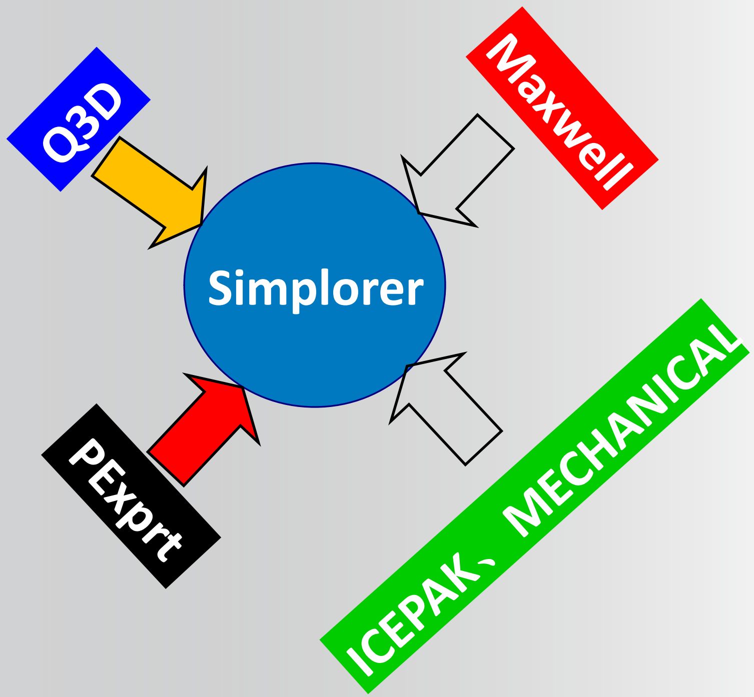

电源系统设计
(主电路拓扑、控制原理)
电磁兼容分析设计
（滤波电路、线缆、传导干扰）
电磁部件设计分析
(变压器、电感、开关频率)
开关器件设计
(IGBT、MOSFET)
电源系统
热、应力设计
可靠性设计
...

# ANSYS电源系统设计仿真平台方案

ANSYS

→ 为工程师提供一个从部件到系统、从基本性能到复杂电磁兼容、从电磁设计到其它多物理域设计的多层次一体化平台




PExprt – 变压器和电感专家设计工具；

Maxwell – 部件电磁特性有限元分析工具；

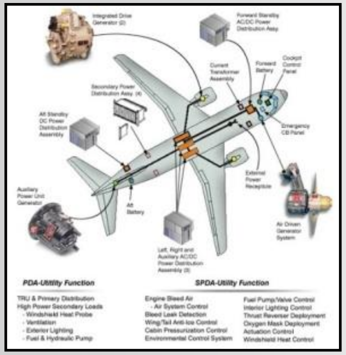
Q3D Extractor – 提取PCB走线、接头及线缆的寄生参数，是进行EMI EMC仿真的重要工具；

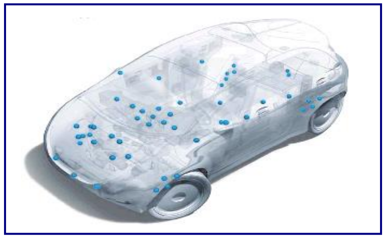
Simplorer – IGBT、MOSFET参数化建模、电源系统整体仿真分析

ANSYS 其它物理域软件 – 如ICEPAK、MECHANICAL、FLUENT,
分析电源系统热、应力等特性

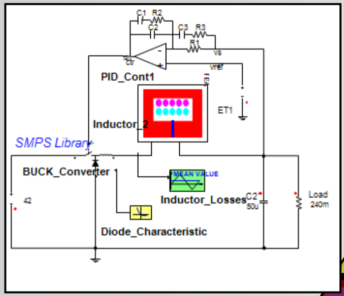
# ANSYS电源系统设计仿真平台技术优势

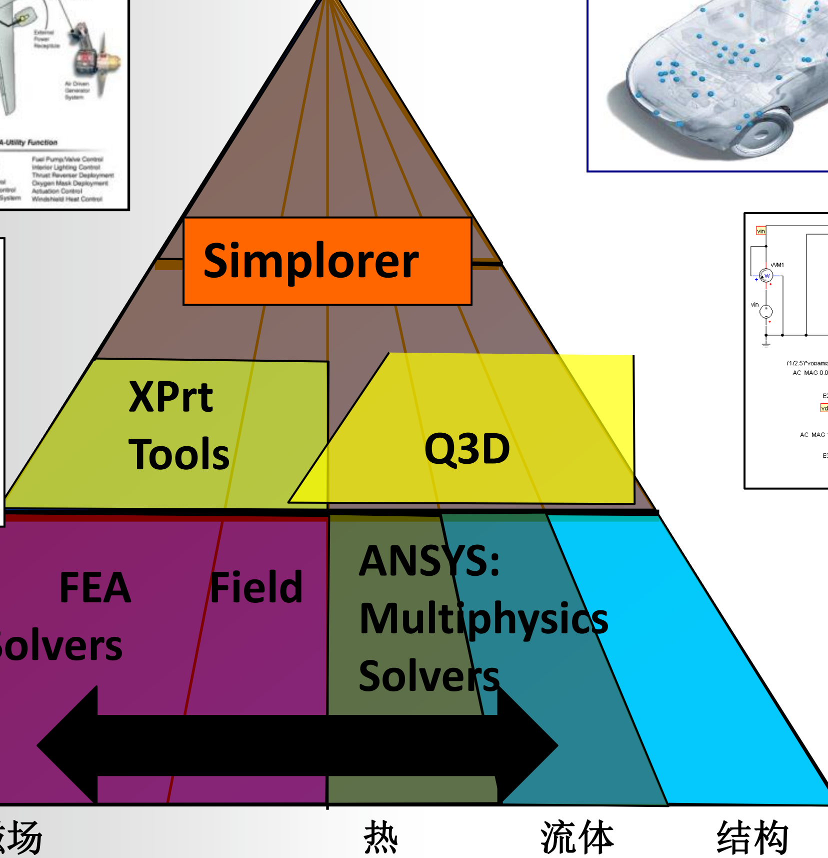
系统集成

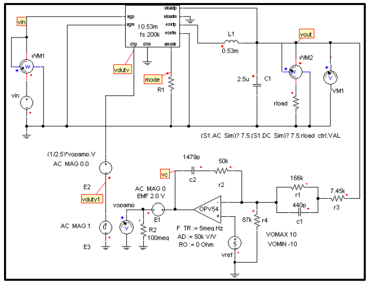




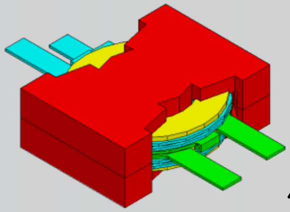
控制算法、
子系统

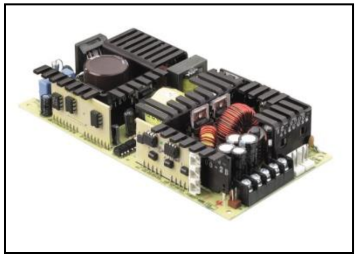


部件





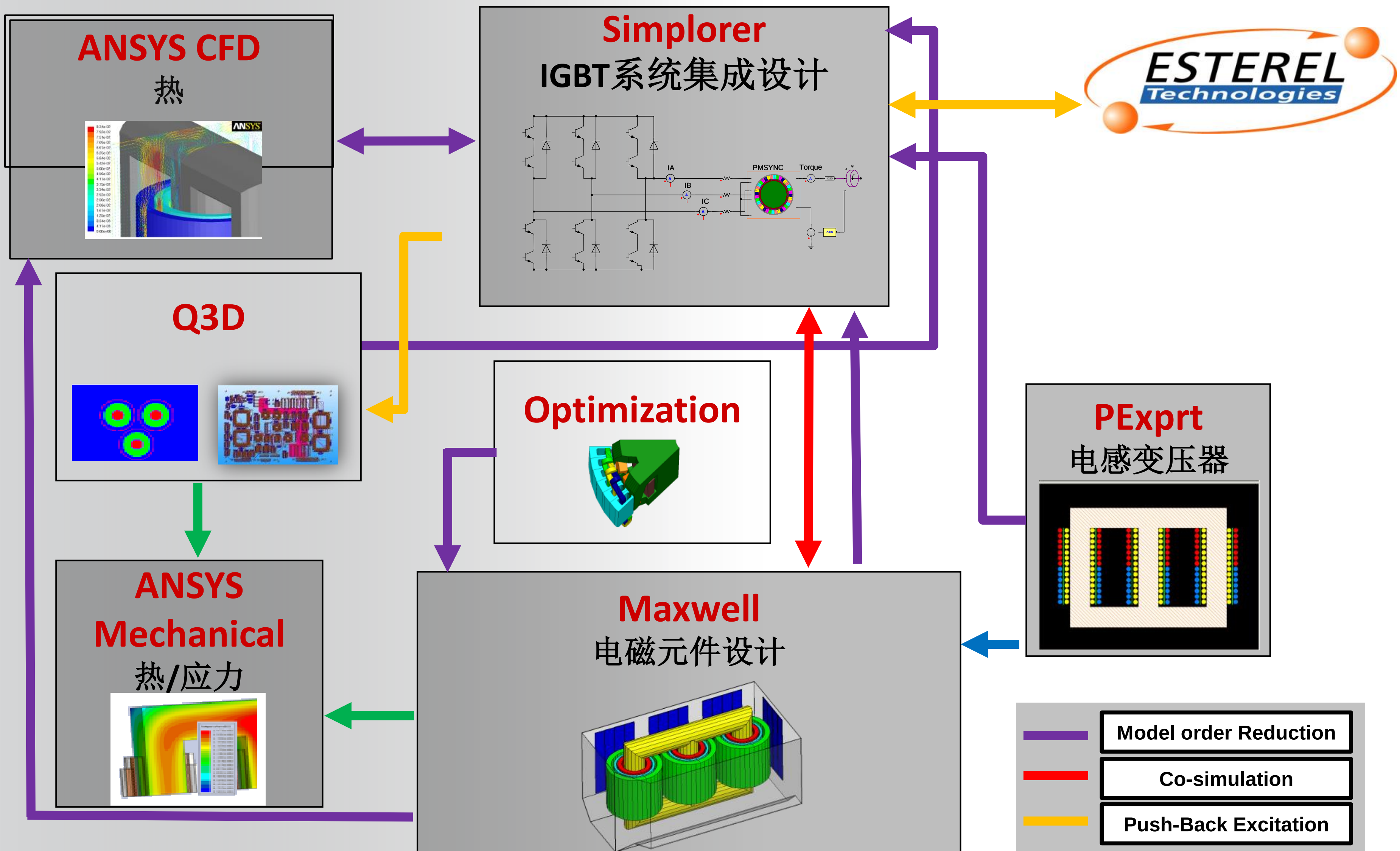
# 内容提要

典型电源系统及其挑战

ANSYS电源系统设计平台实现

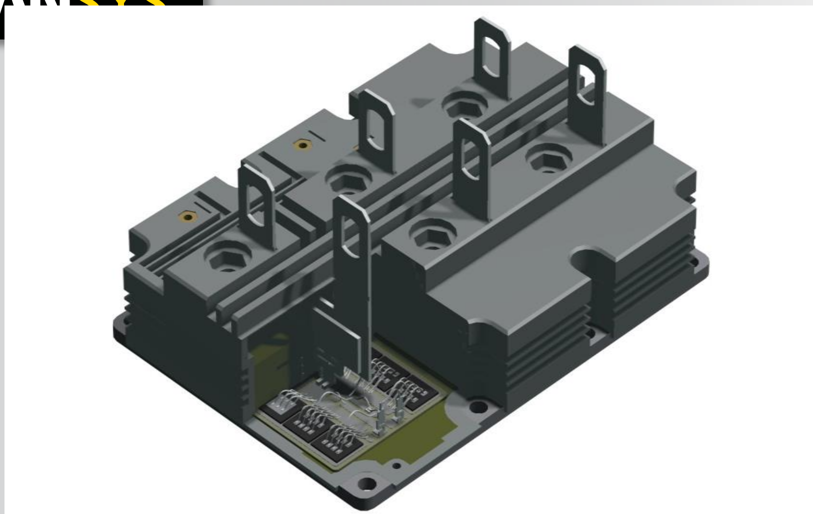
总结

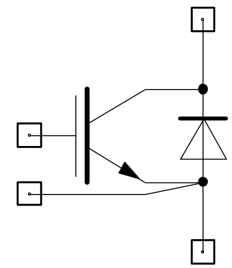
# 电源系统在ANSYS平台上的设计流程



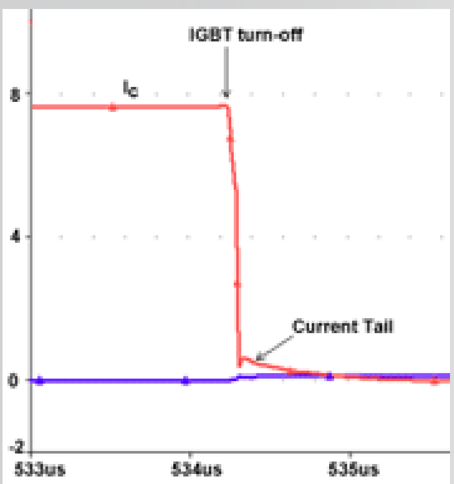

  

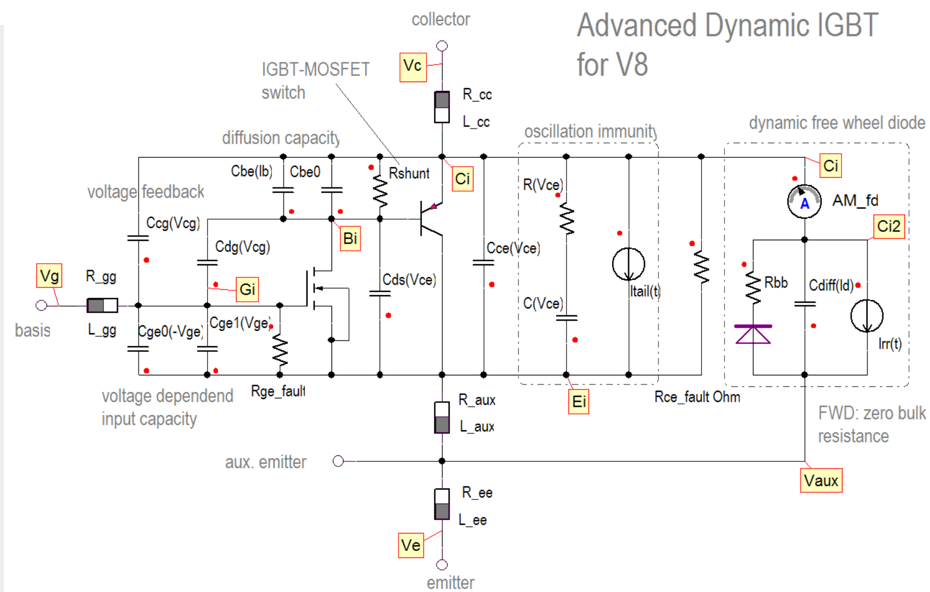
# 开关电源器件建模(IGBT、MOSFET)





NIGBT_AdvDyn1

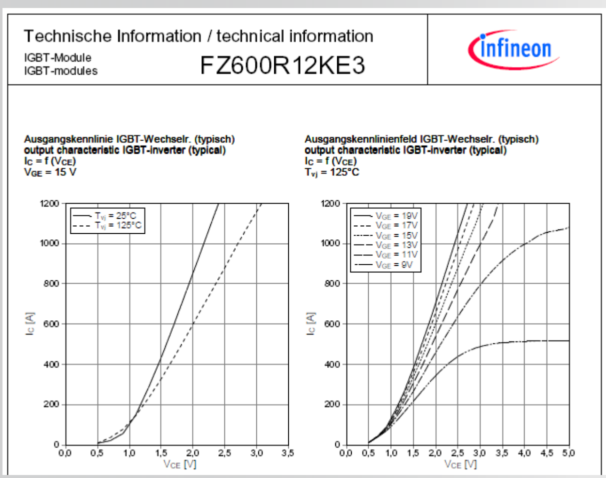


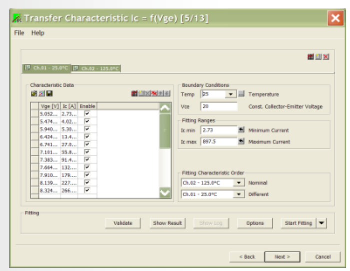

**Simplorer中IGBT参数化建模能体现其工作细节**

# Simplorer IGBT、MOSFET特征化建模

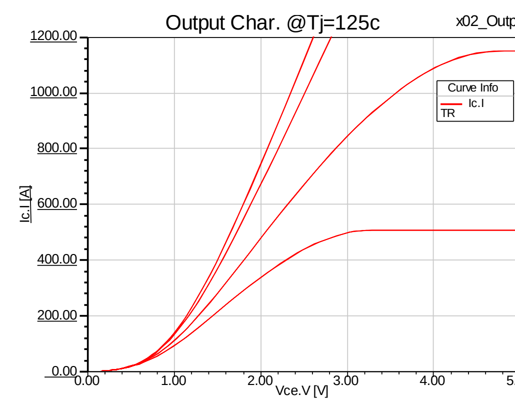
Infineon :
eupec FZ600R12KE3


数据手册数据

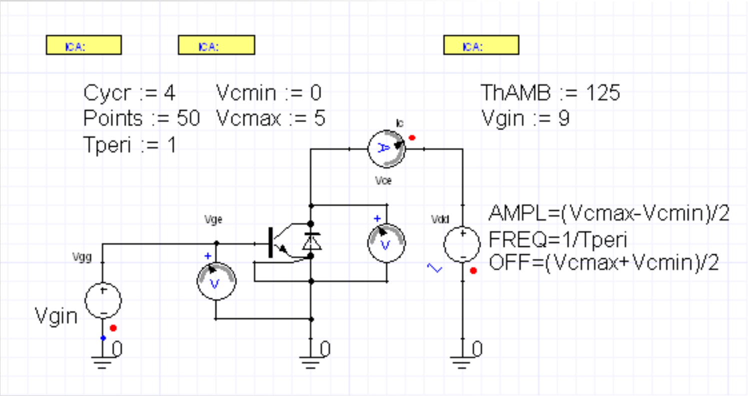


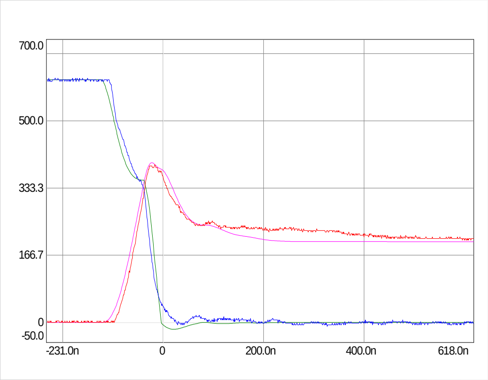

提取工具

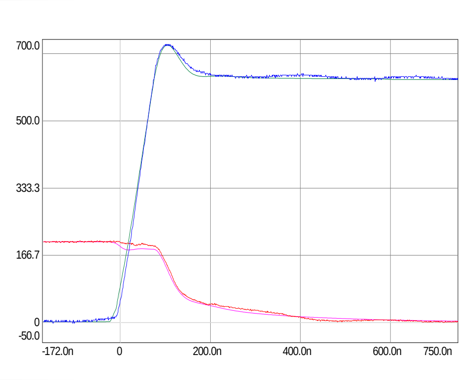



输出特性 Vce-Ic

# Simplorer IGBT、MOSFET特征化建模

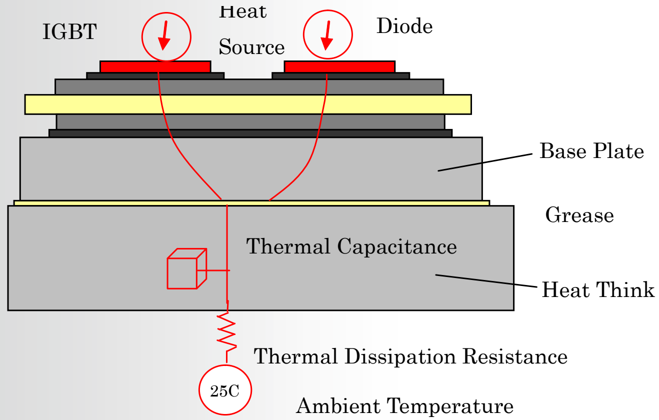


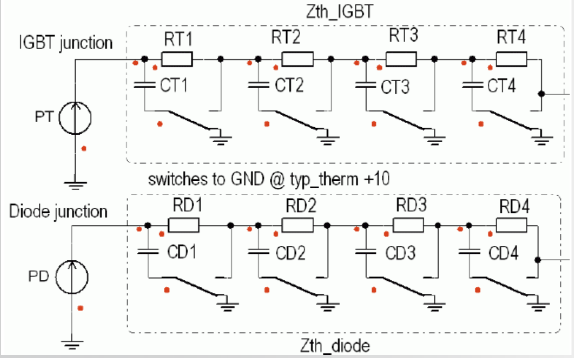



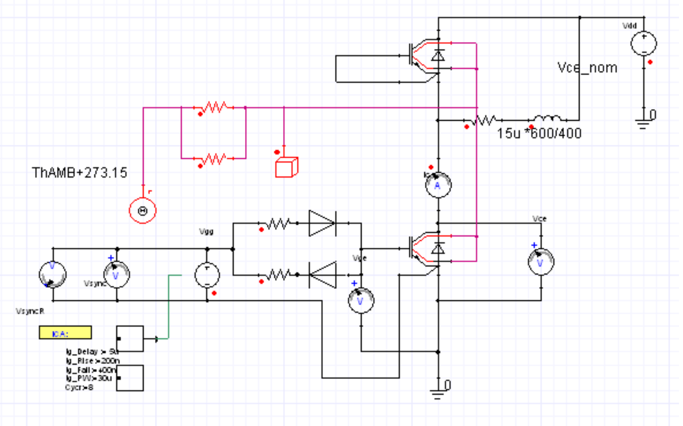

**测试 VS 仿真(Vce and Ic)**

# Simplorer IGBT、MOSFET特征化建模



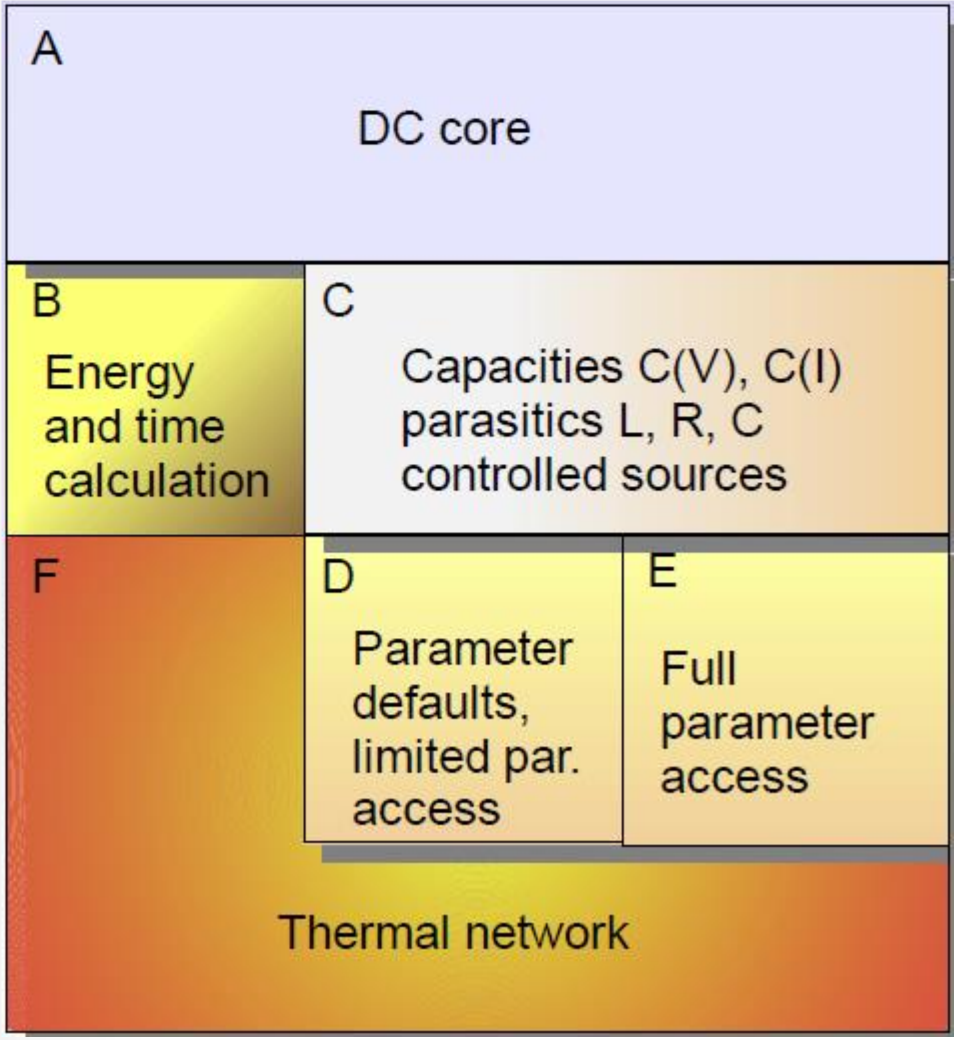

热网络
（IGBT & Diode）




IGBT发热仿真

# Simplorer中IGBT参数化模型的层次

- 平均模型:
  - A, B, F
- 基本动态模型:
  - A, C, D, F
- 高级动态模型 :
  - A, C, E, F



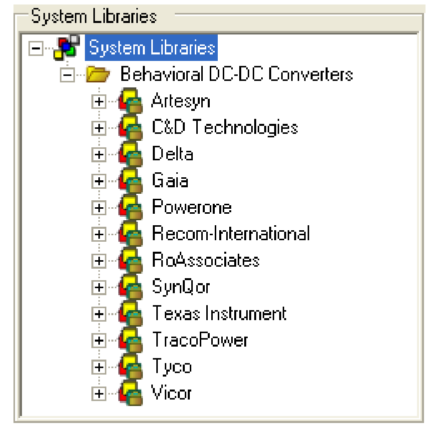
# 开关电源器件建模(IGBT、MOSFET)

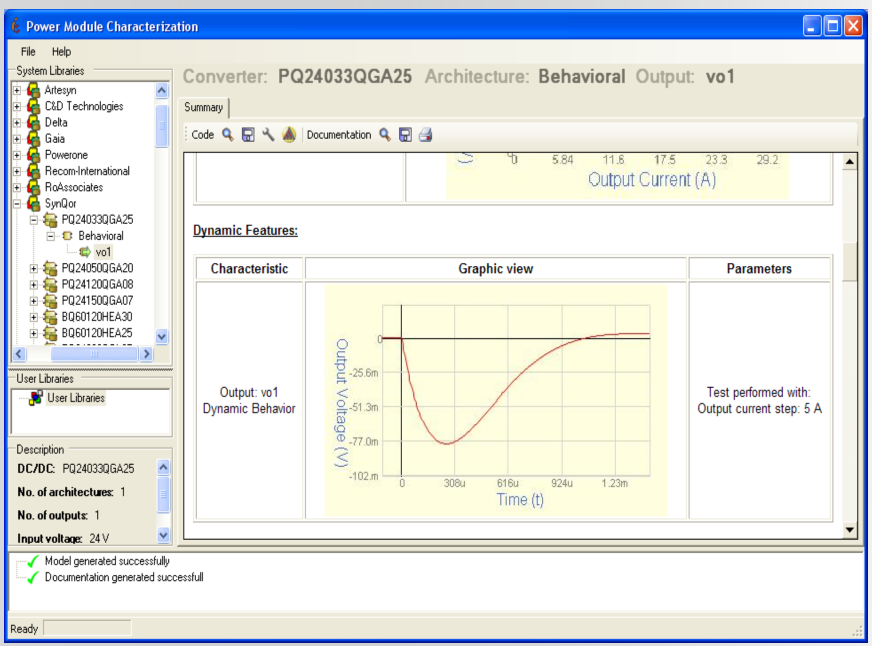
| | | |
|---|---|---|
| 基本电性能和热性能 | → | Average IGBT model |
| 驱动电路优化、EMI/EMC | → | Advanced dynamic model |

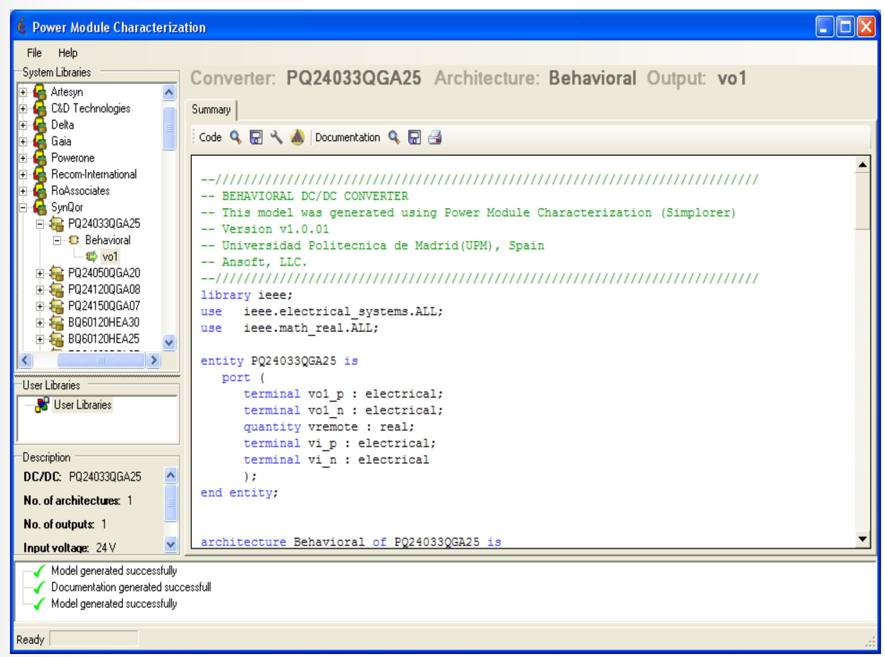
根据仿真需求确定合适的IGBT建模类型

# Simplorer DC/DC特征化建模工具

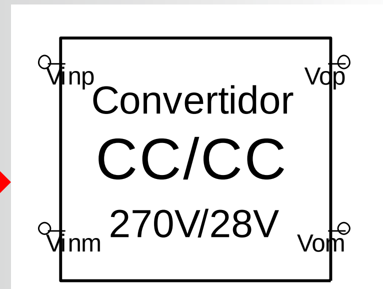
- 根据厂商的数据手册或测试结果建模
- 生成DC/DC行为模型，提高求解速度和效率
- 不用关心DC/DC模块拓扑等内部细节

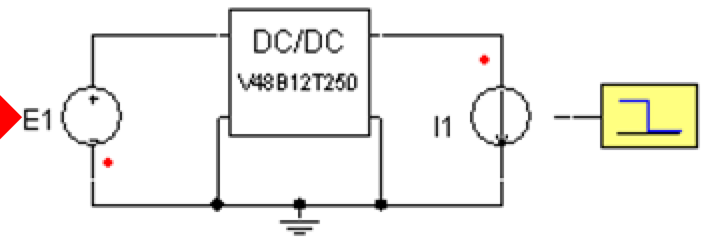


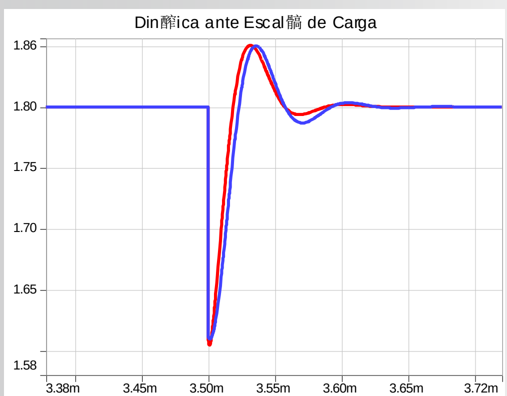



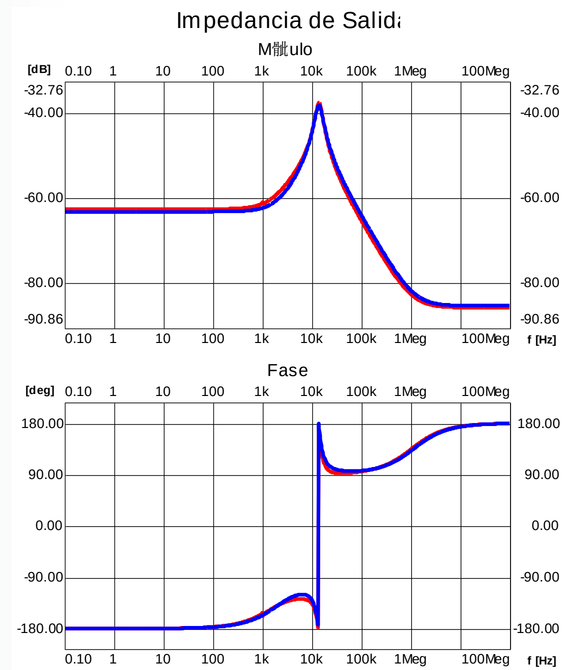



# Simplorer DC/DC特征化建模工具

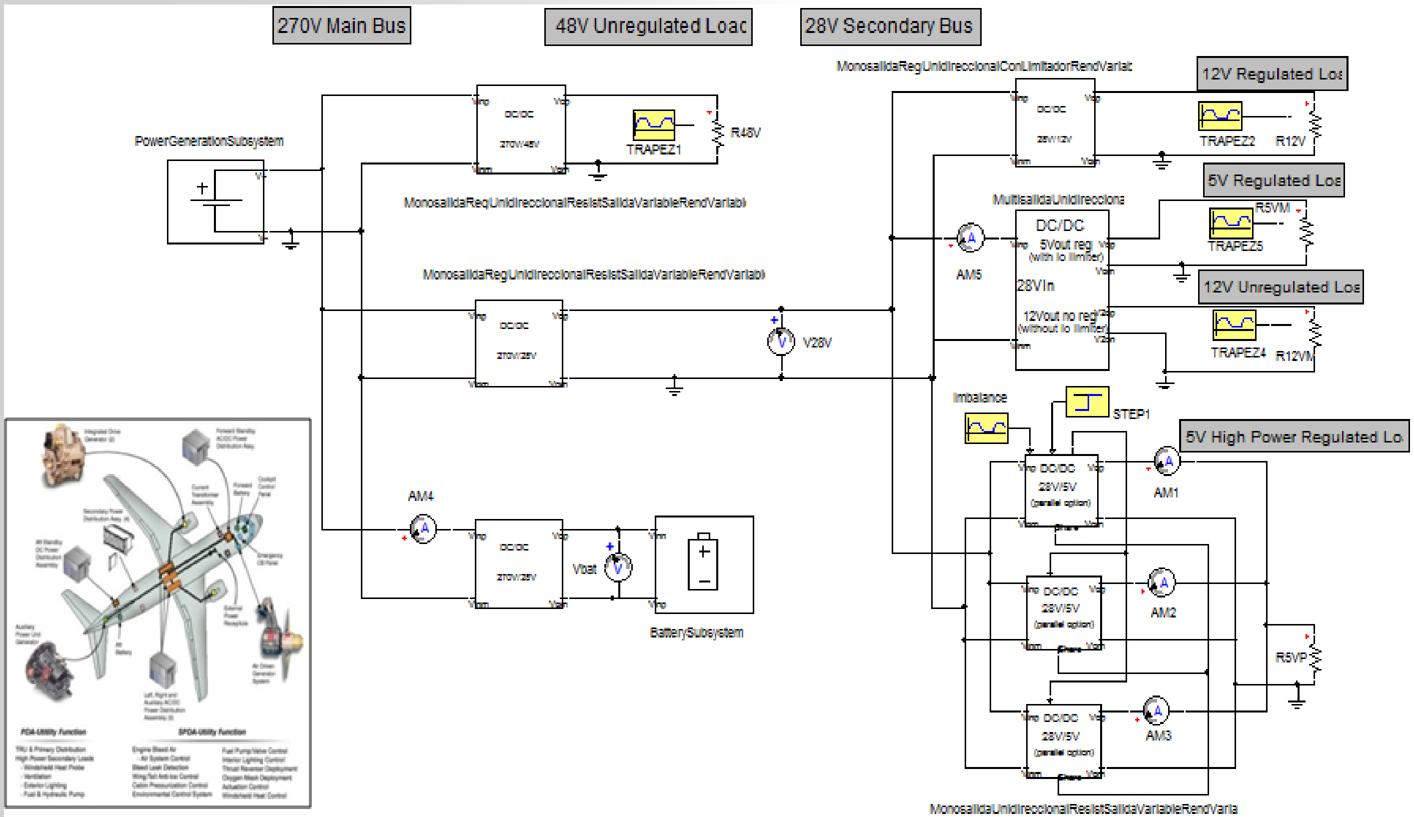



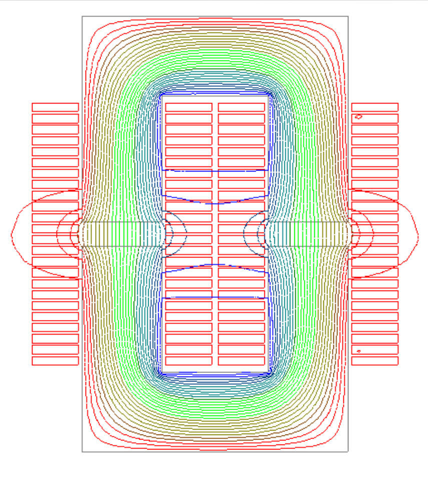

测试 VS 仿真(时域和频域)

# Simplorer DC/DC特征化建模工具




生成DC/DC行为模型，用于大系统仿真，分析系统效率、上电延时逻辑、冲击电流等

# 电磁部件设计建模（变压器，电感器）



激磁电感等基本参数设计

漏感、耦合电容等寄生参数

损耗、温升

体积、高度、封装

........................

# 电磁部件设计建模（变压器，电感器）

PExprt: 变压器和电感电磁专家设计和优化工具

包含典型器件生产厂家模型库:

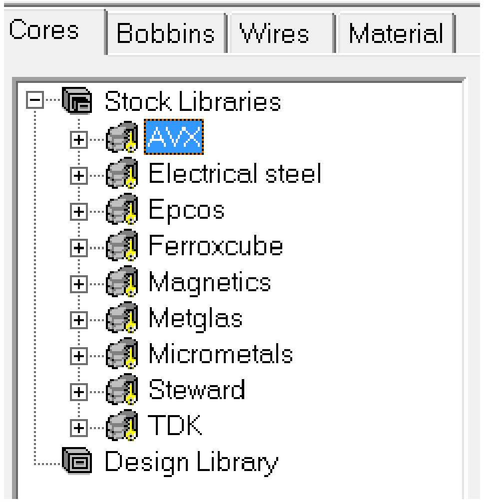
- 铁芯, 骨架, 导线
- 环形, 平面, 线绕

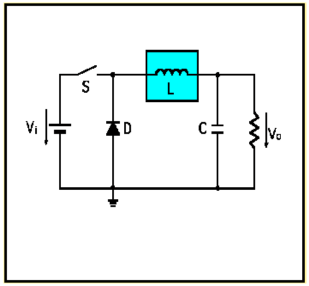
考虑集肤效应、临近效应、气隙效应和热效应

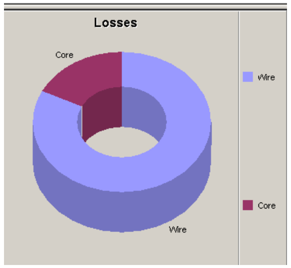
考虑绕组损耗、铁芯损耗、R,L,C参数和温升

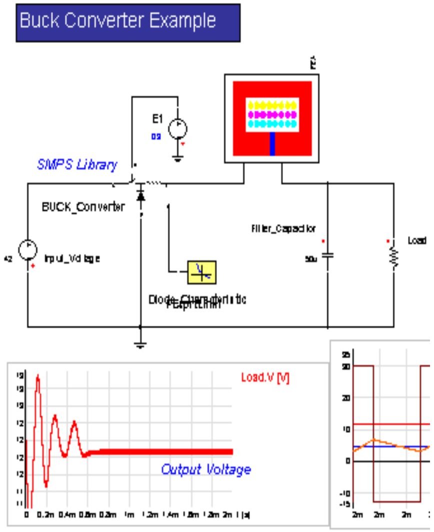
可以与MAXWELL无缝集成进行更细致电磁场分析

可以集成到Simplorer中进行系统仿真



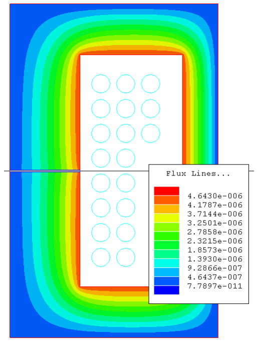



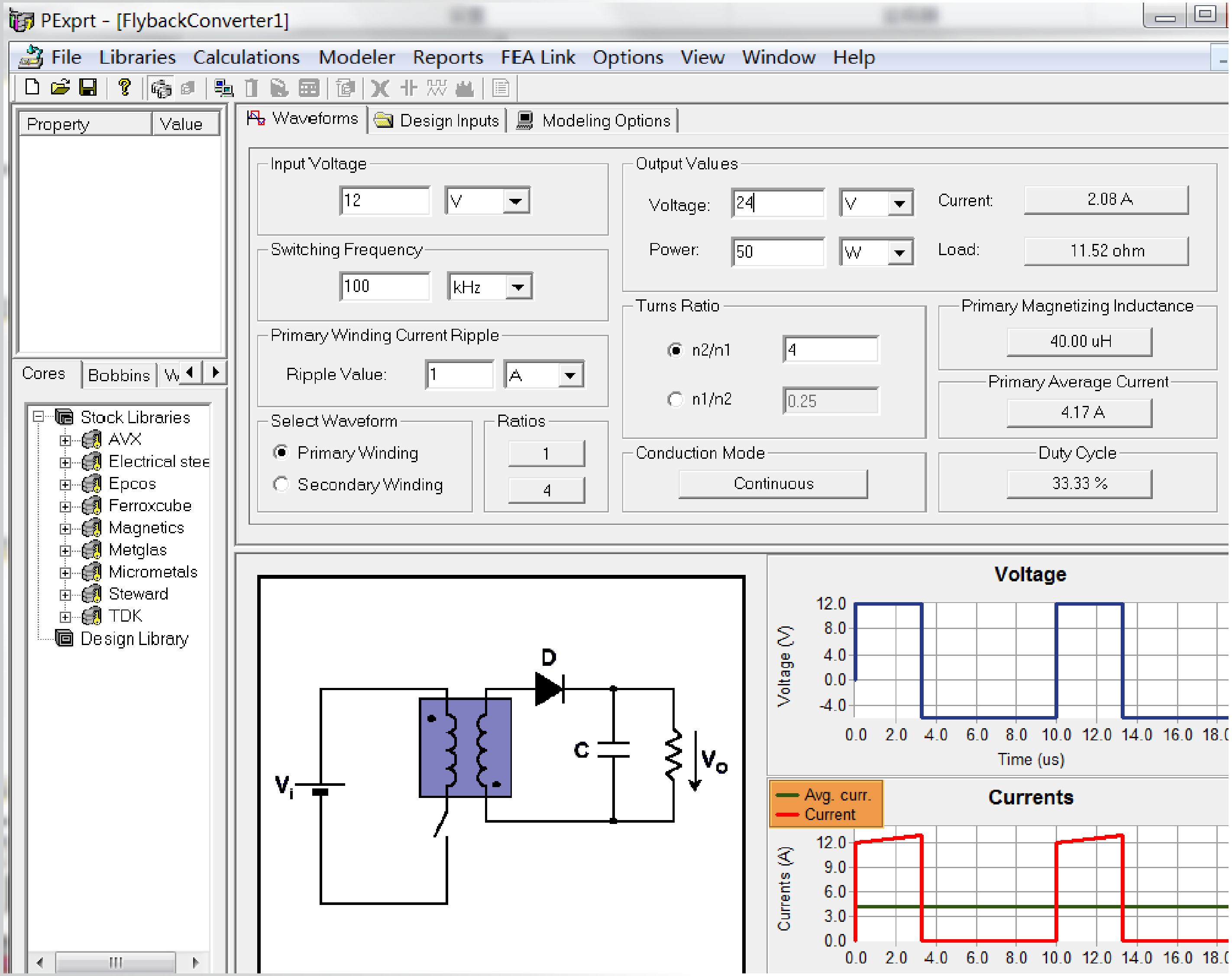

SIMPLORER

Maxwell



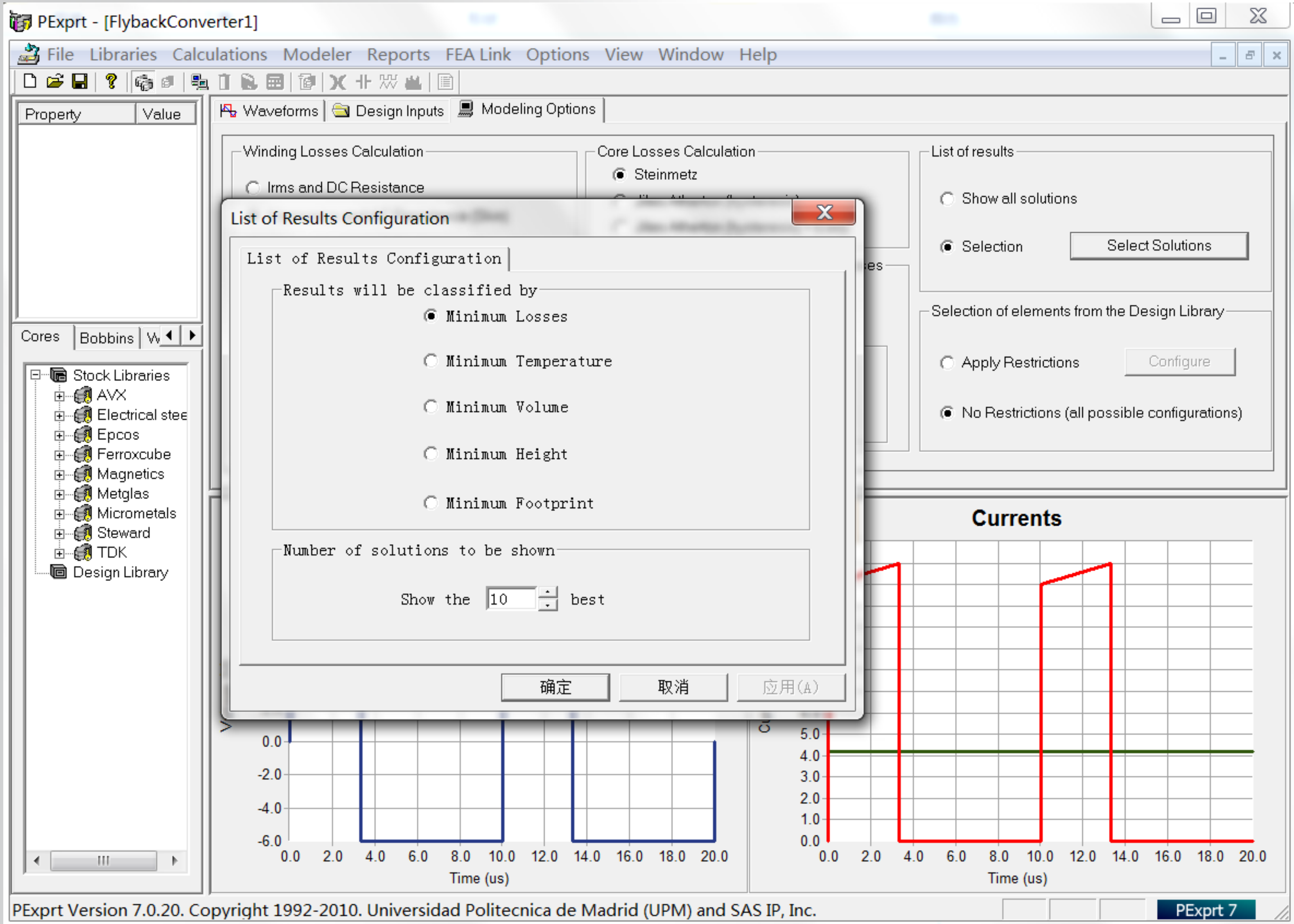
  

# PExprt 设计流程-->指定性能指标

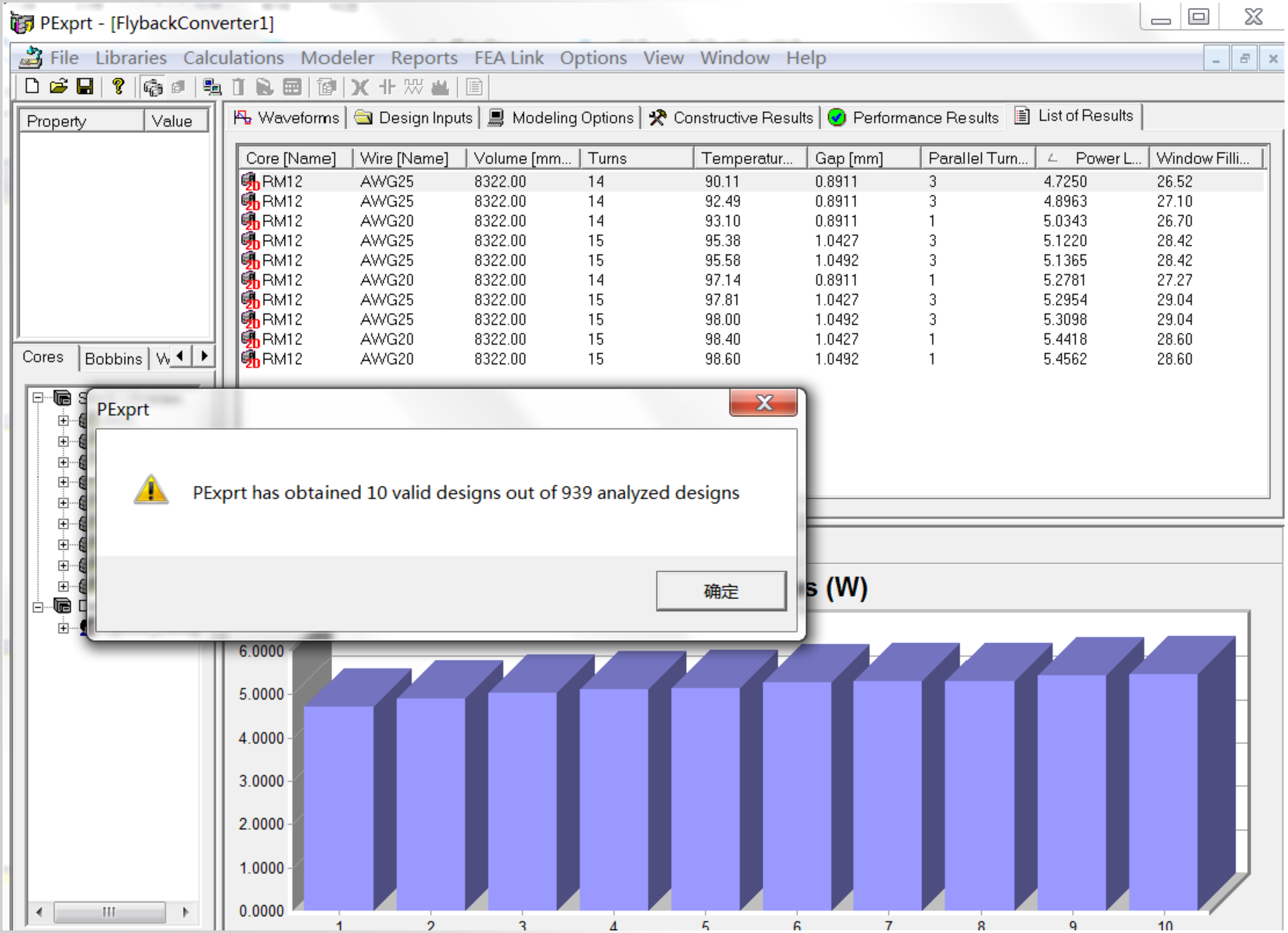


# PExprt 设计流程-->指定设计目标

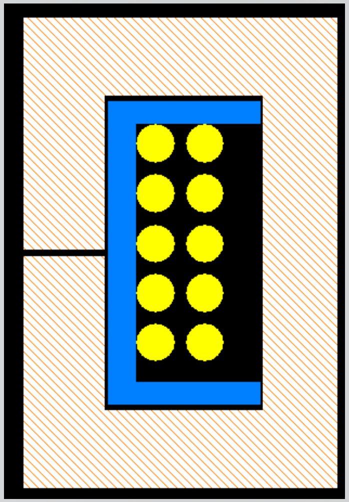


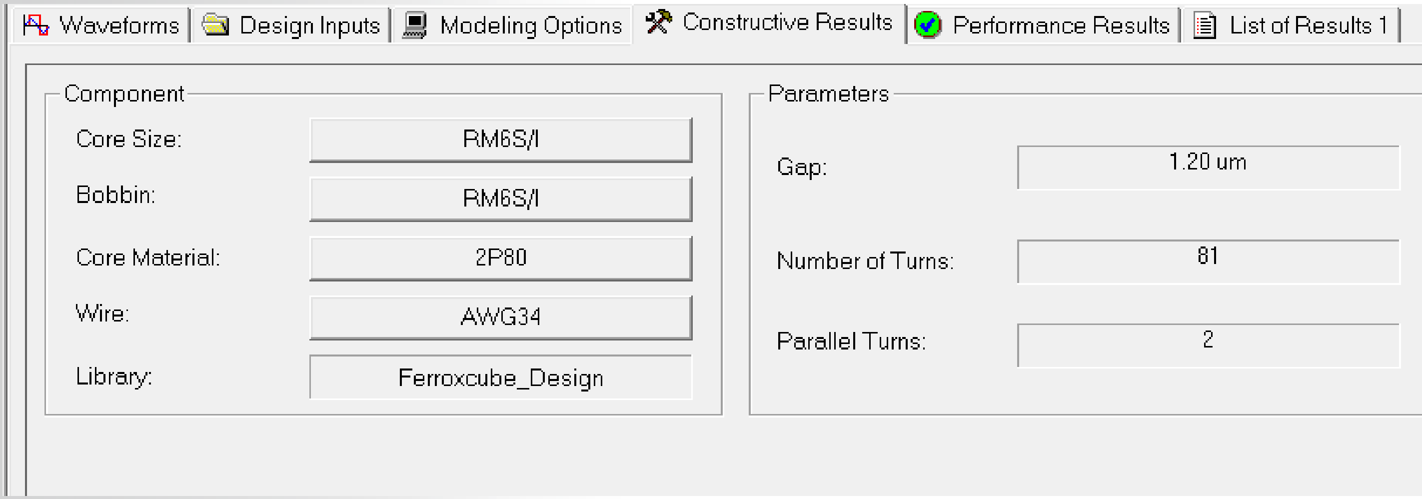
# PExprt 设计流程-->快速得到结果

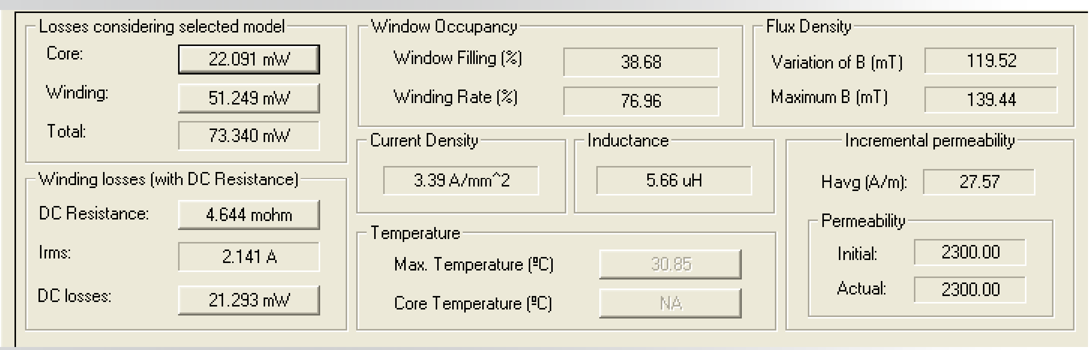


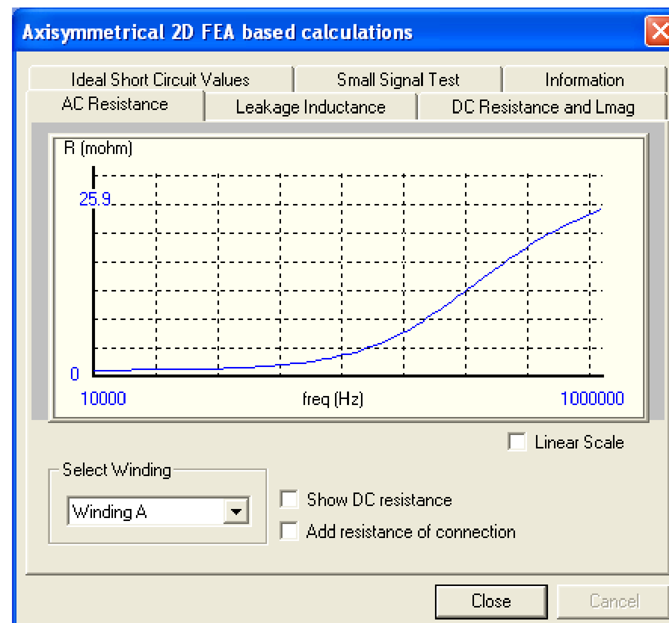

  May 31, 2013

# PExprt设计流程-->结果信息丰富





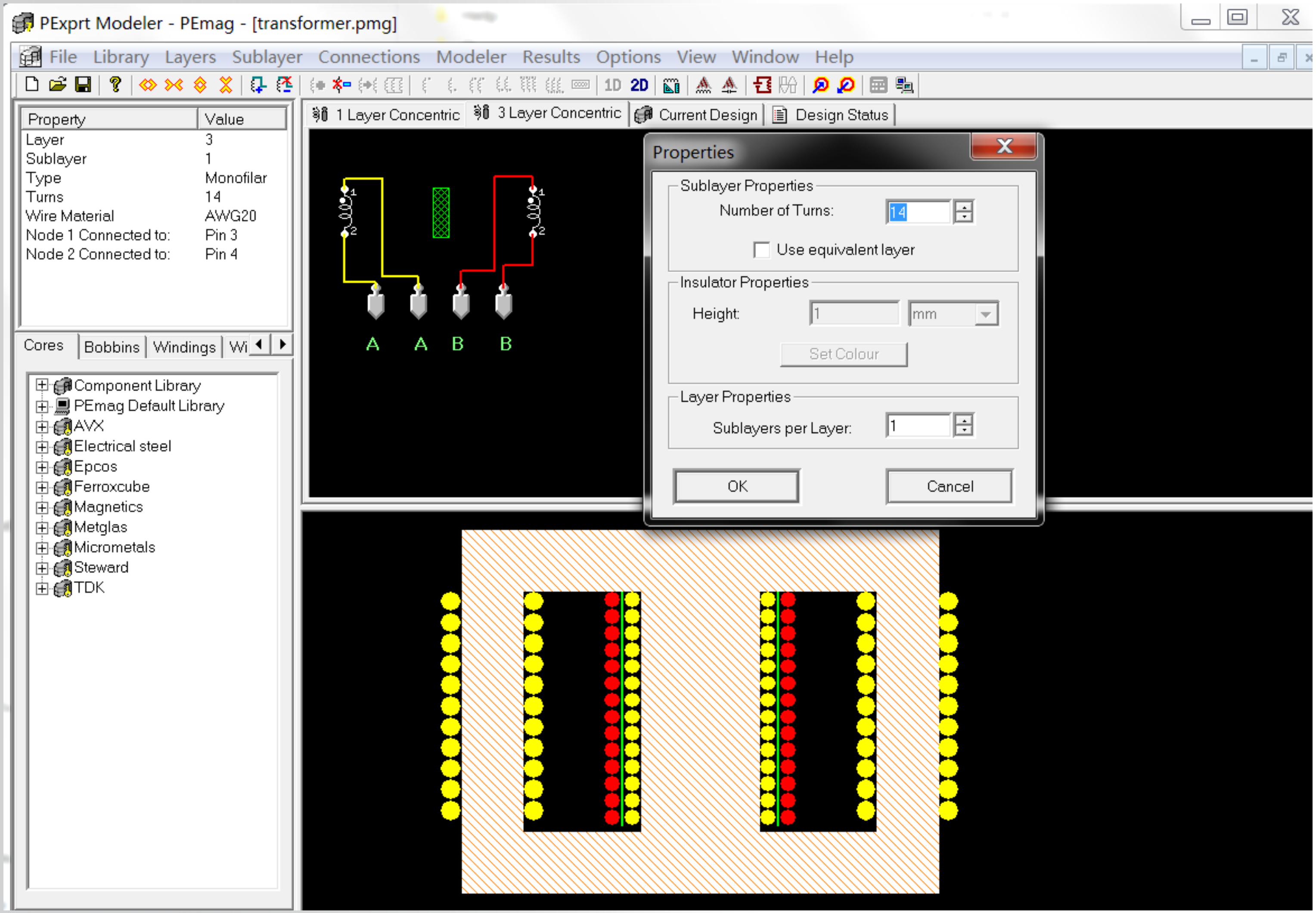



ANSYS

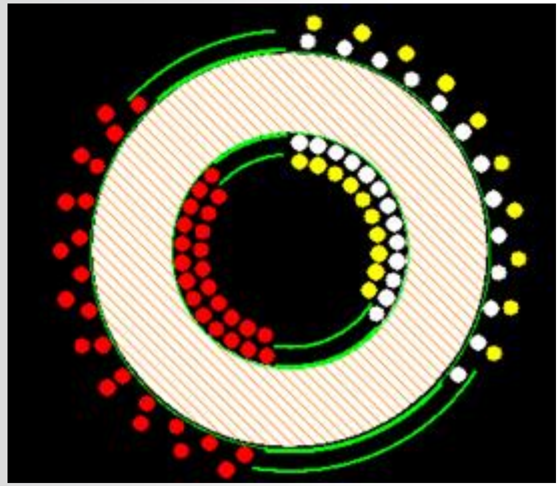
# PExprt

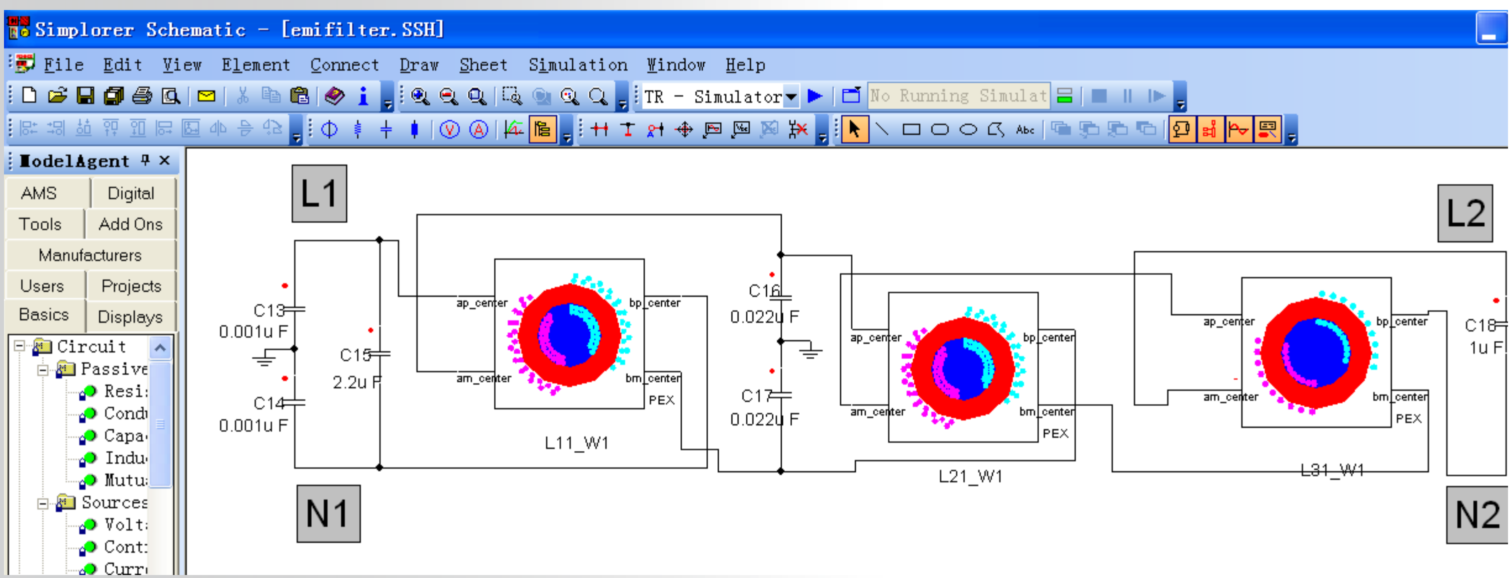
直接用PEmag设计、分析、优化变压器或电感



# PExprt

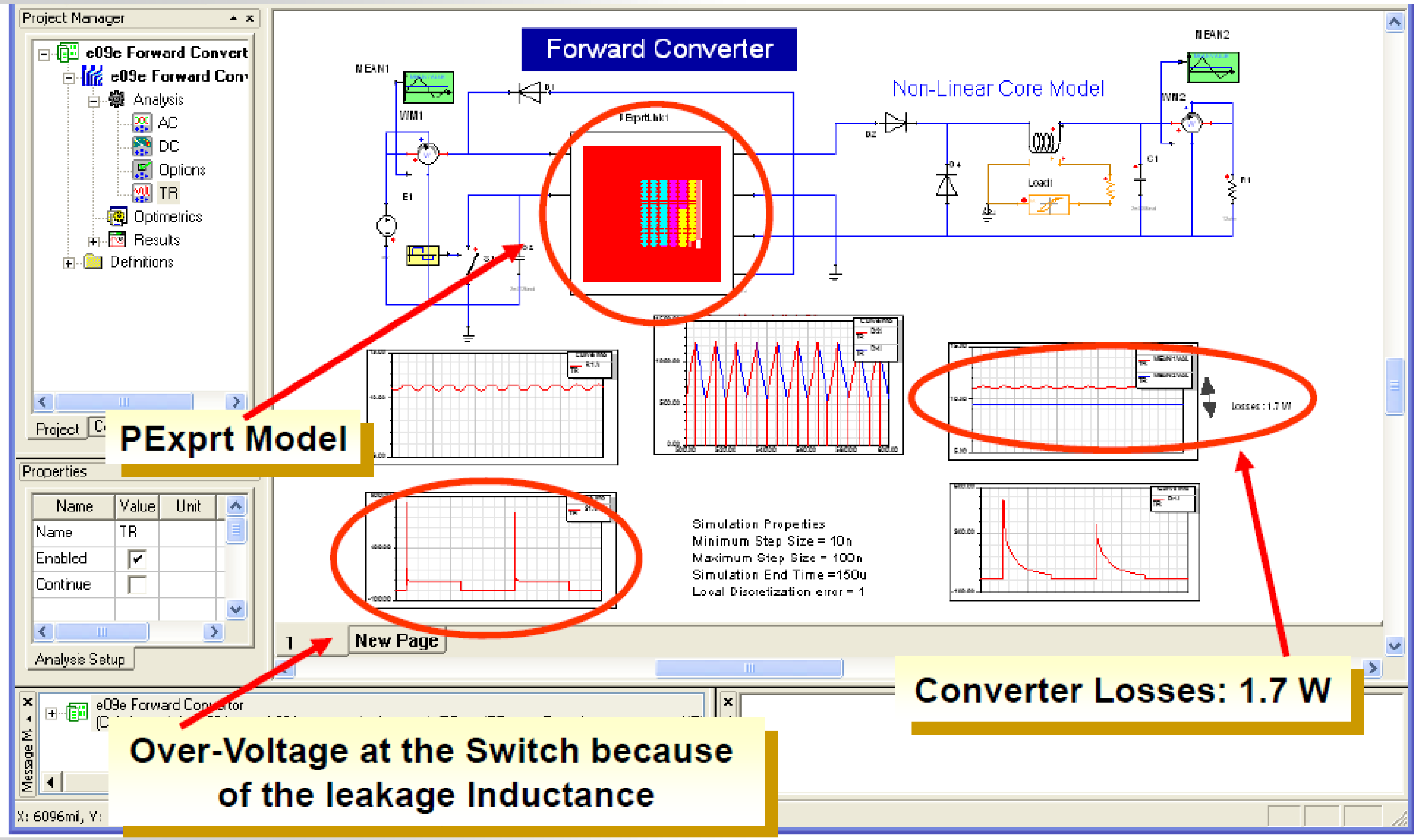
直接用PEmag设计差模、共模滤波电感






  May 31, 2013

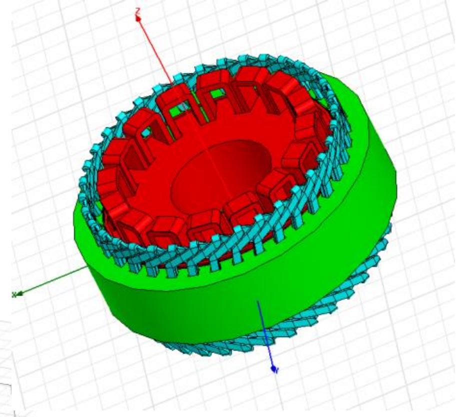
# 典型的PExprt设计案例

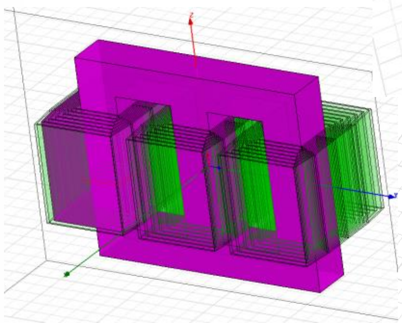


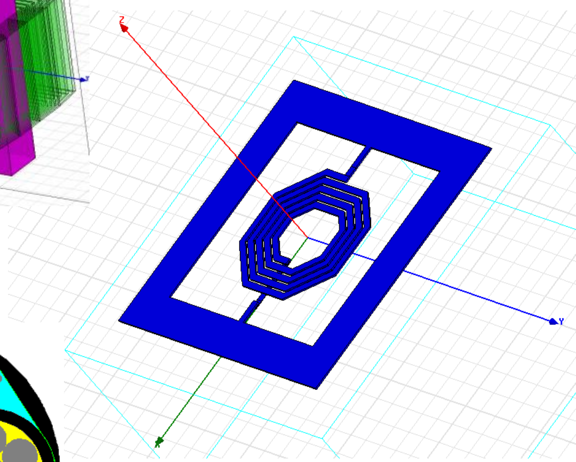
# 电磁部件设计建模（变压器，电感器）

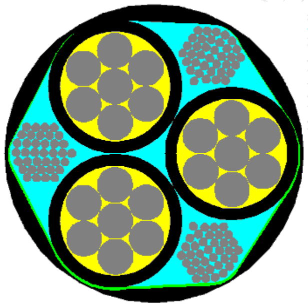
## Maxwell

- 用FEA方法解决 2-D 、3-D 电磁场问题
- 分析 R,L,C, 力, 转矩, 损耗,饱和效应, 时变效应等
- 仿真:电感, 变压器, 电动机,发电机 , 执行器, 母排等
- 与Simplorer协同仿真
- 与PExprt
- 与ANSYS Mechanical直接链接



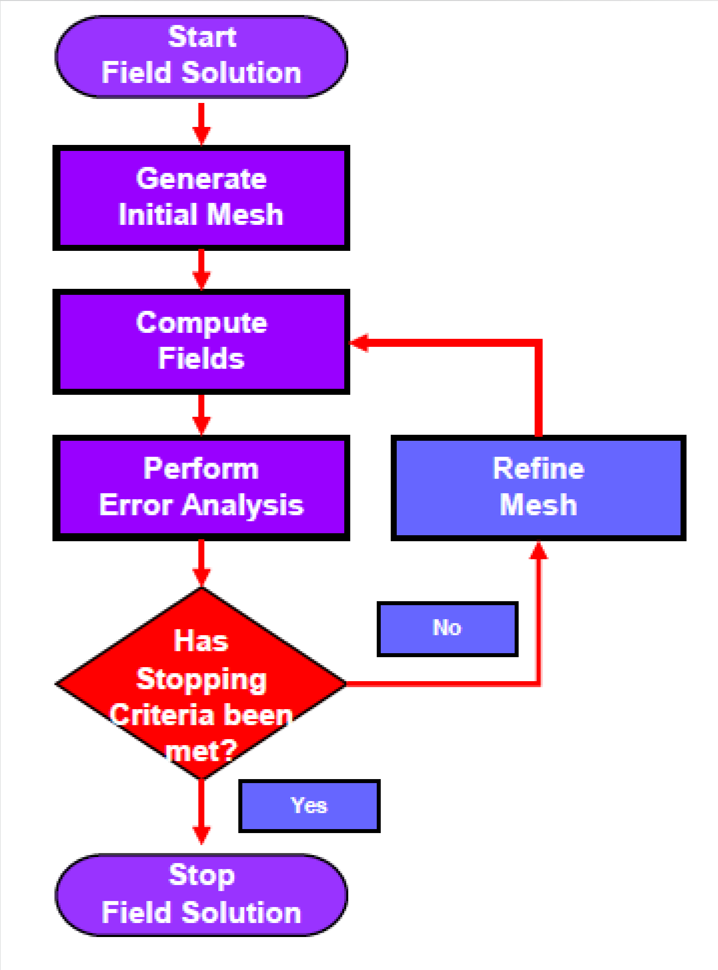


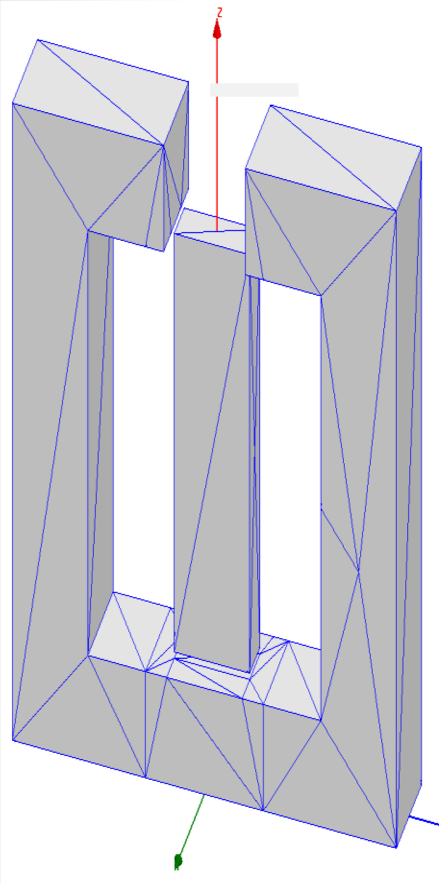


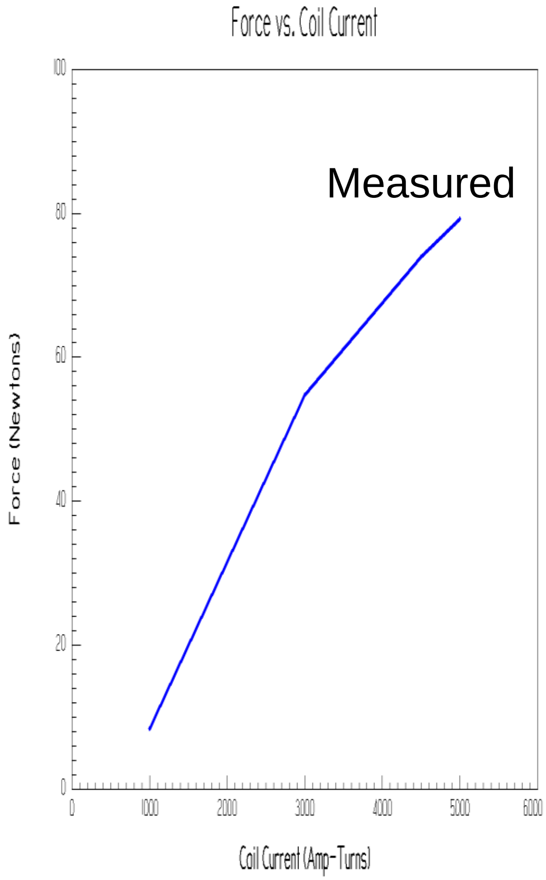


# 自动自适应网格剖分

**Ansoft核心技术之一，FEA计算精度的保证**

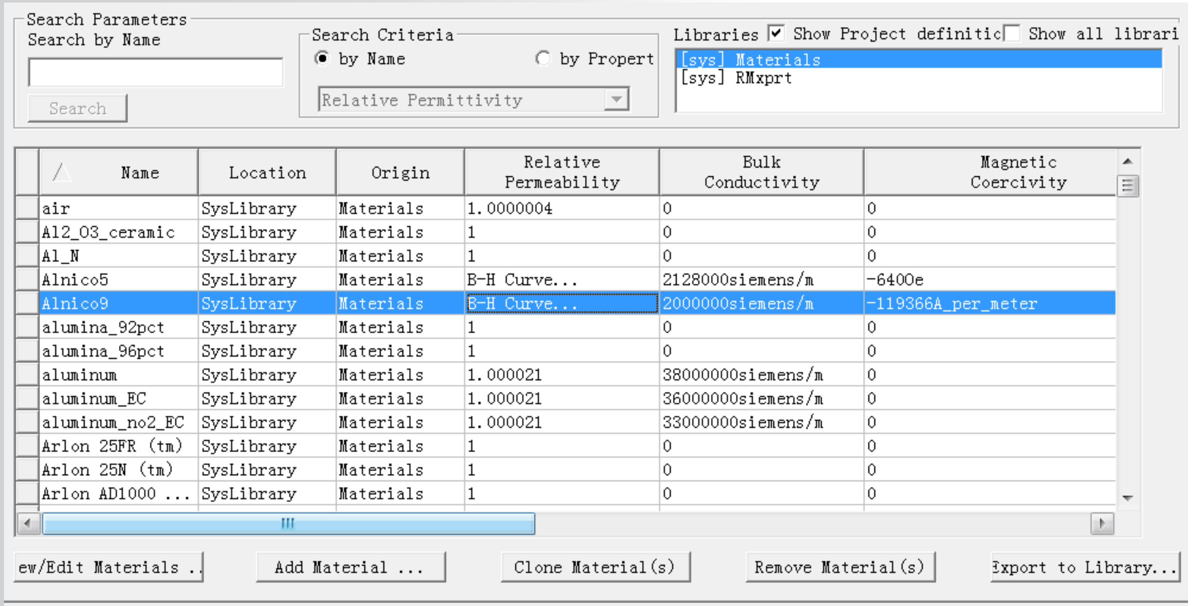


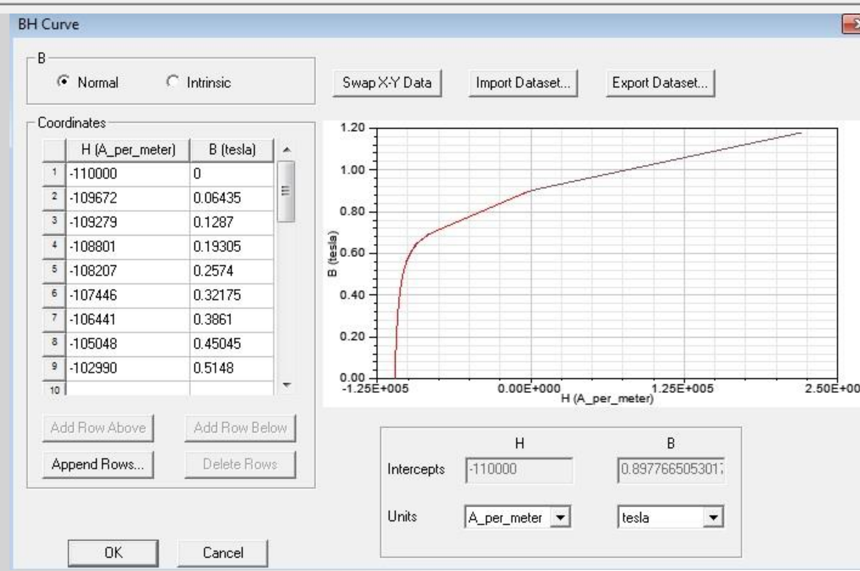



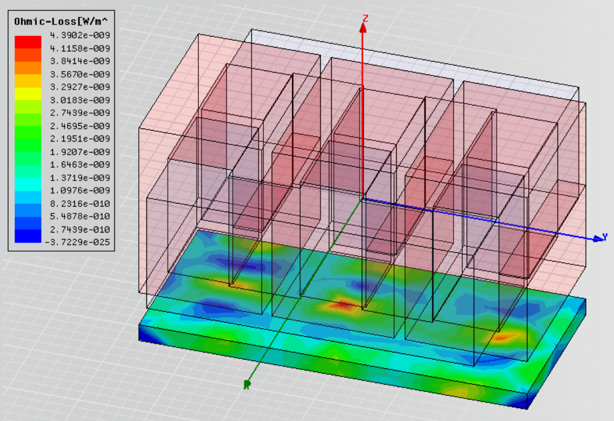
# Maxwell材料参数

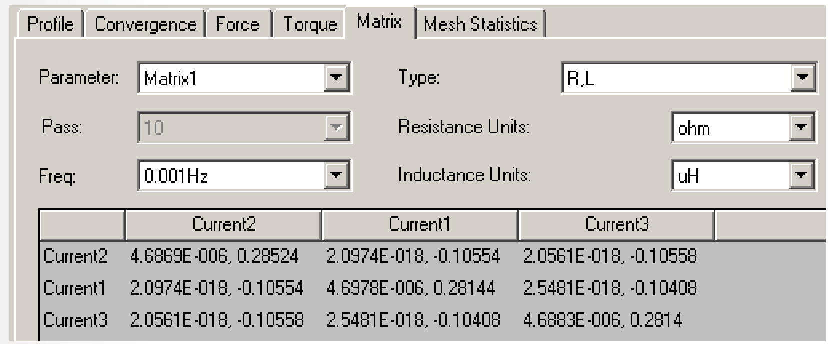


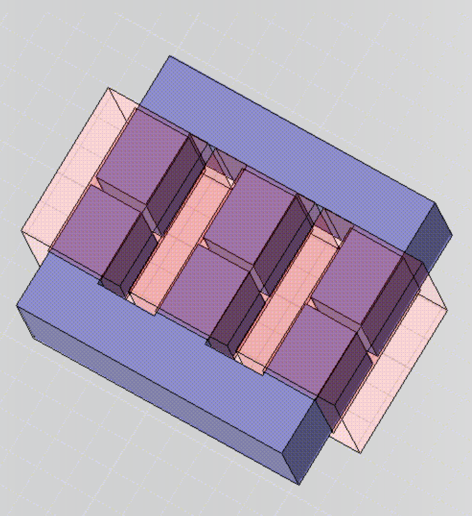
# Maxwell电感计算

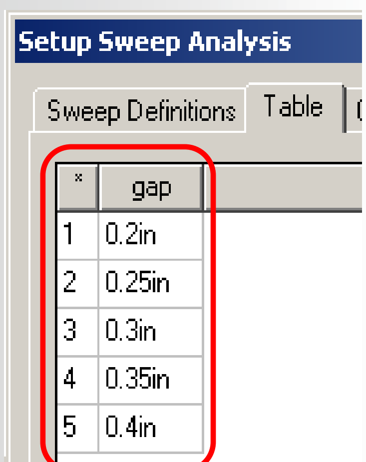
- 考虑电感边缘效应、非线性饱和、频率影响等



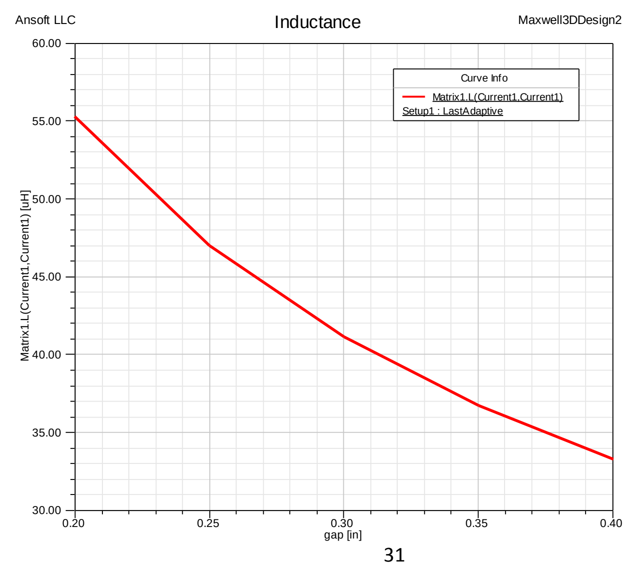




| | Current2 | Current1 | Current3 |
|---|---|---|---|
| Current2 | 4.6869E-006, 0.28524 | 2.0974E-018, -0.10554 | 2.0561E-018, -0.10558 |
| Current1 | 2.0974E-018, -0.10554 | 4.6978E-006, 0.28144 | 2.5481E-018, -0.10408 |
| Current3 | 2.0561E-018, -0.10558 | 2.5481E-018, -0.10408 | 4.6883E-006, 0.2814 |

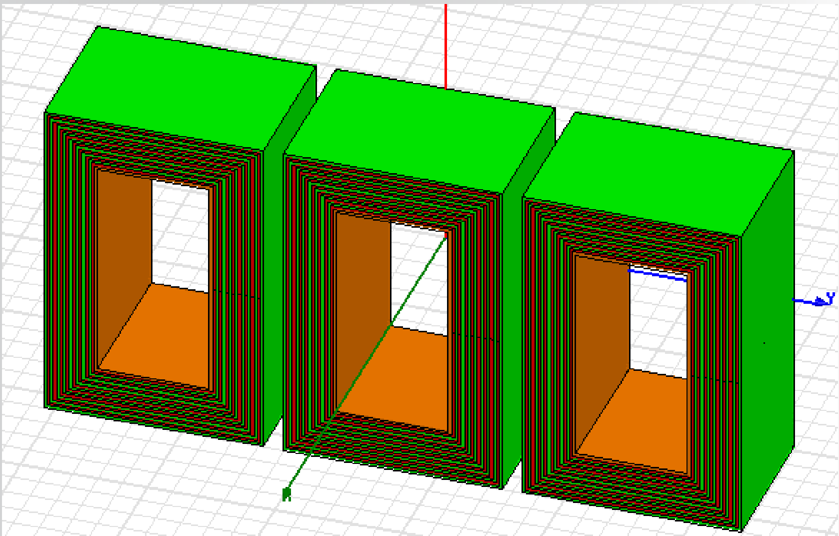


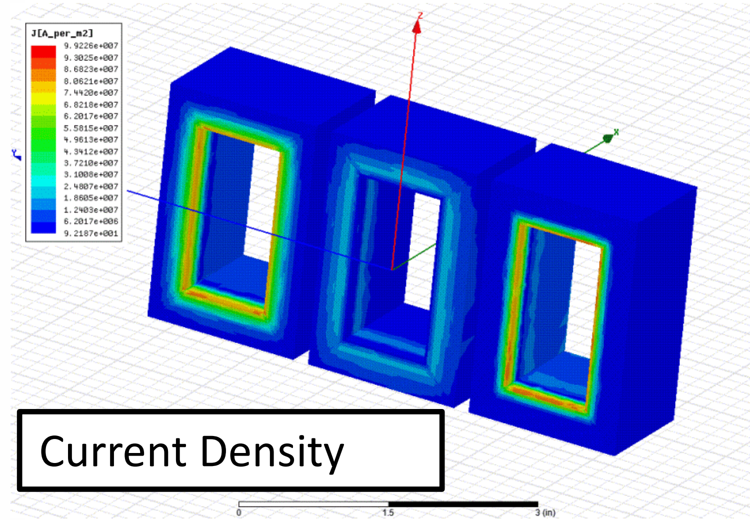


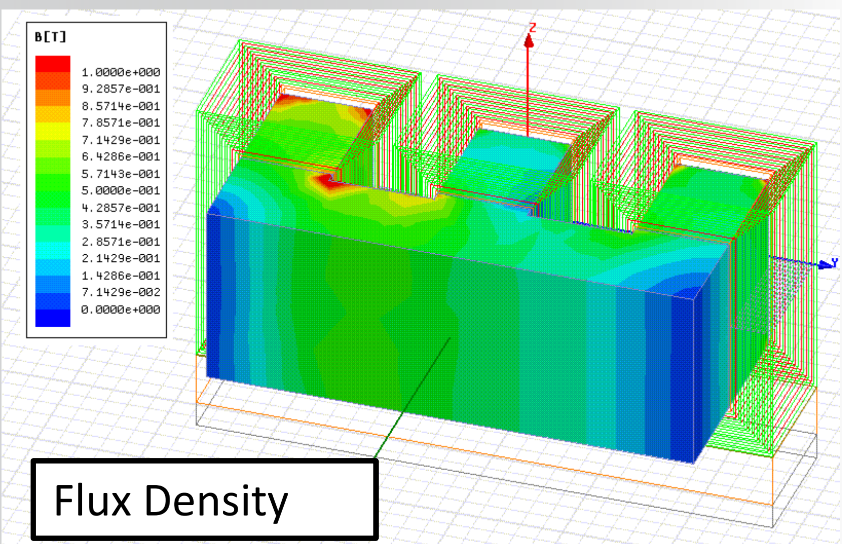

| * | gap |
|---|---|
| 1 | 0.2in |
| 2 | 0.25in |
| 3 | 0.3in |
| 4 | 0.35in |
| 5 | 0.4in |

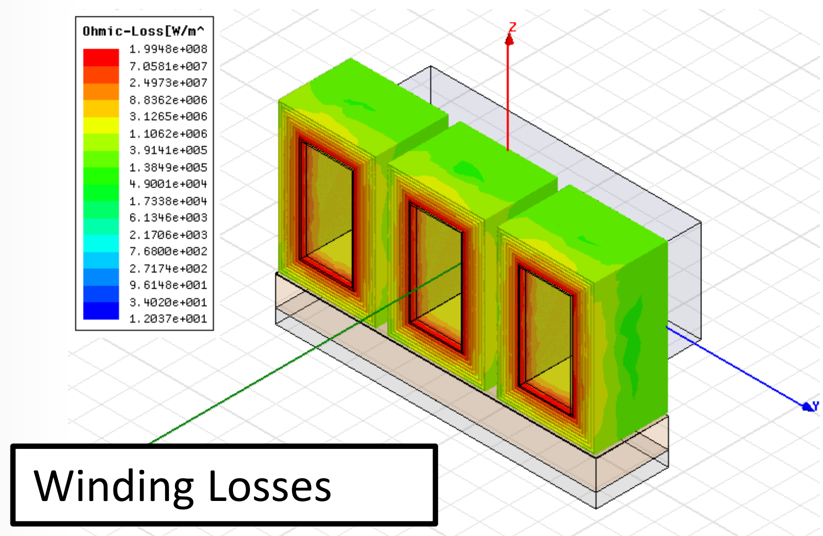


# Maxwell全面分析变压器电感的场数据





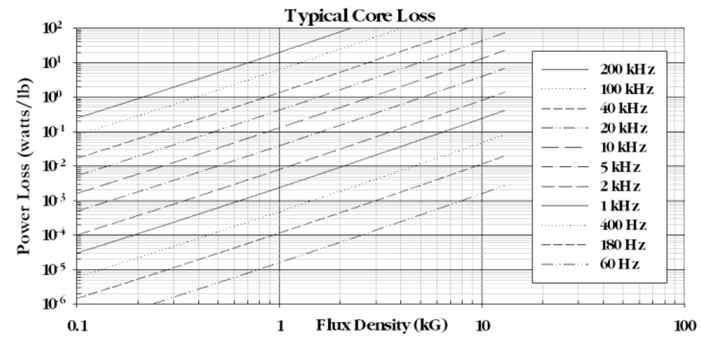



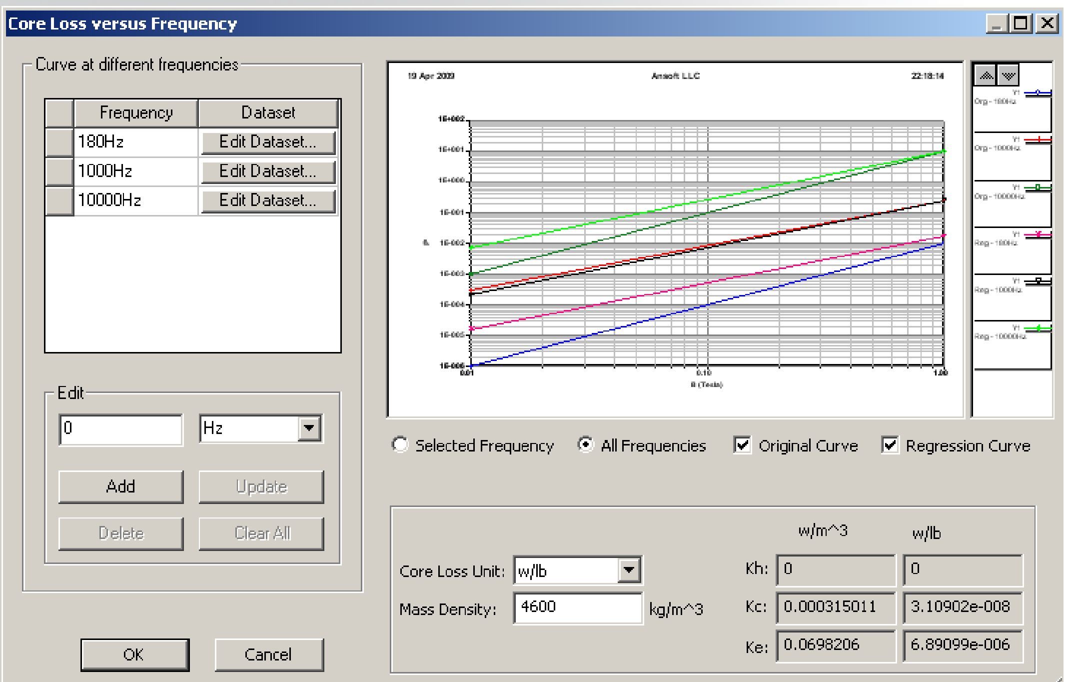



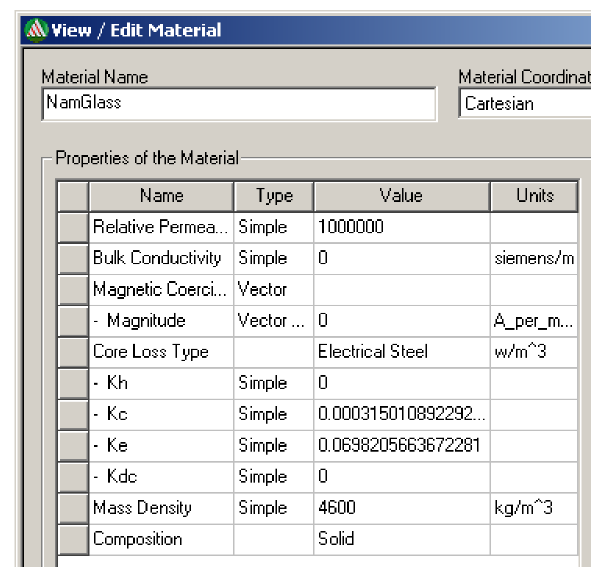

  May 31, 2013

# Maxwell铁芯损耗计算

- 铁芯损耗很大程度上决定电源温升
- 根据铁芯生产厂家提供的数据，输入MAXWELL中，计算铁芯损耗，可以是单频率、多频率（部分材料自带损耗数据）
- 自动提取铁芯的损耗系数



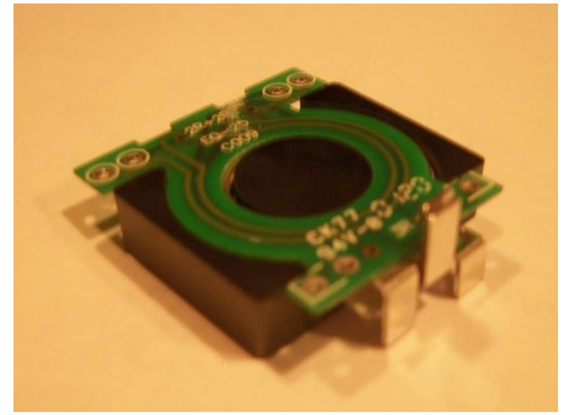



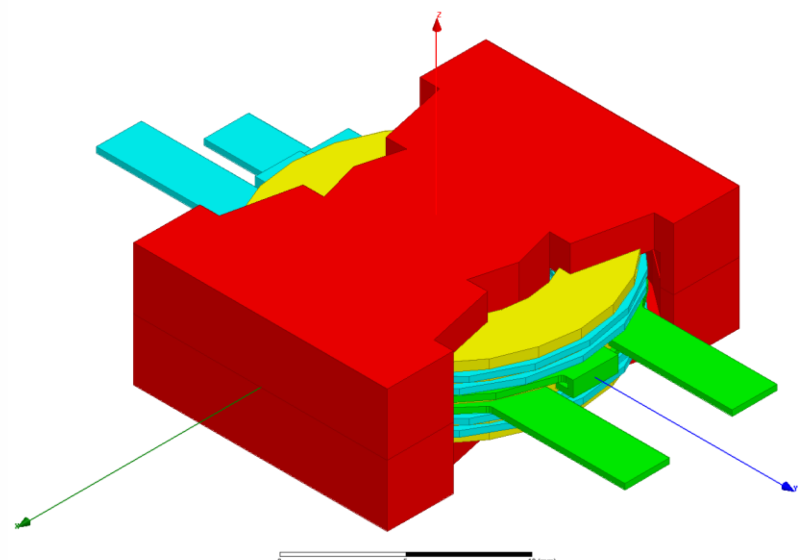



# Maxwell：平板变压器案例

**非标磁性元器件：平板变压器**

- Ferrite PQ Core
- Primary turns = 4
- Secondary turns = 2
- Insulation layers between conductors
- Fundamental Frequency = 100kHz



# 铁氧体材料属性

随频率变化的磁导率实部和磁导率虚部

利用Simplorer的sheetscan功能提取datasheet数据

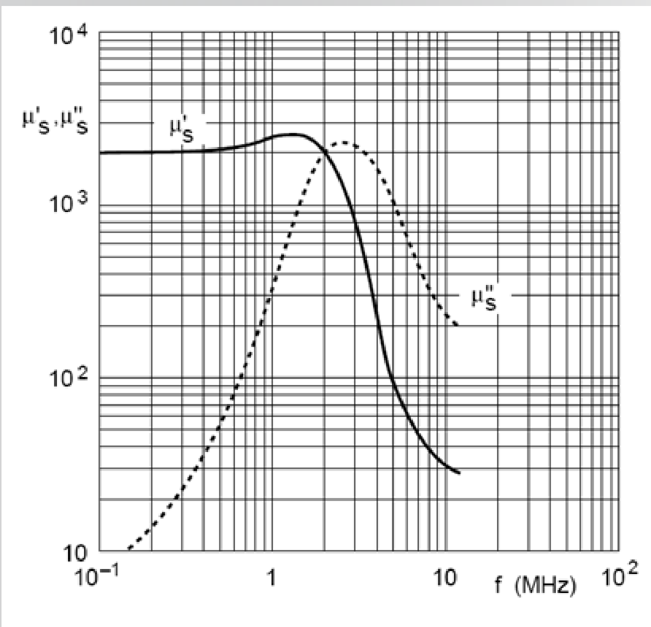
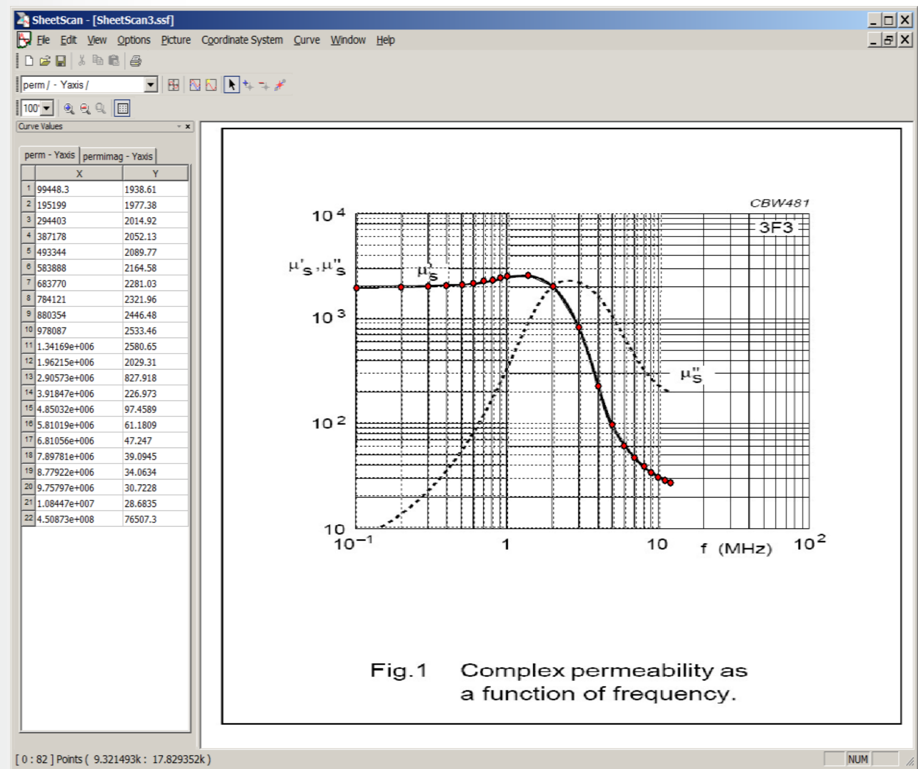



Fig.1 Complex permeability as a function of frequency.

Datasheet




Sheetscan

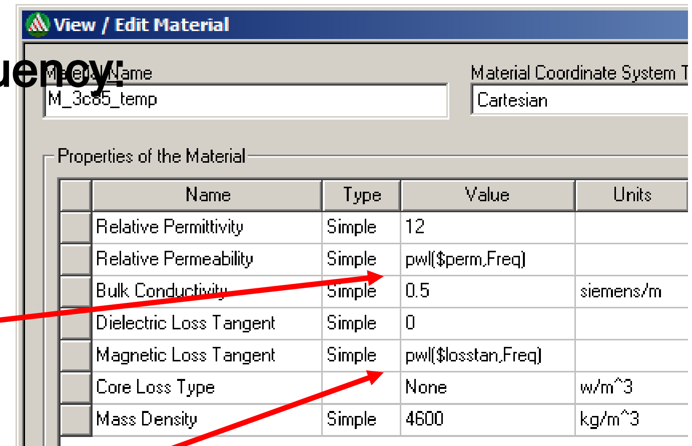
# 铁氧体材料属性

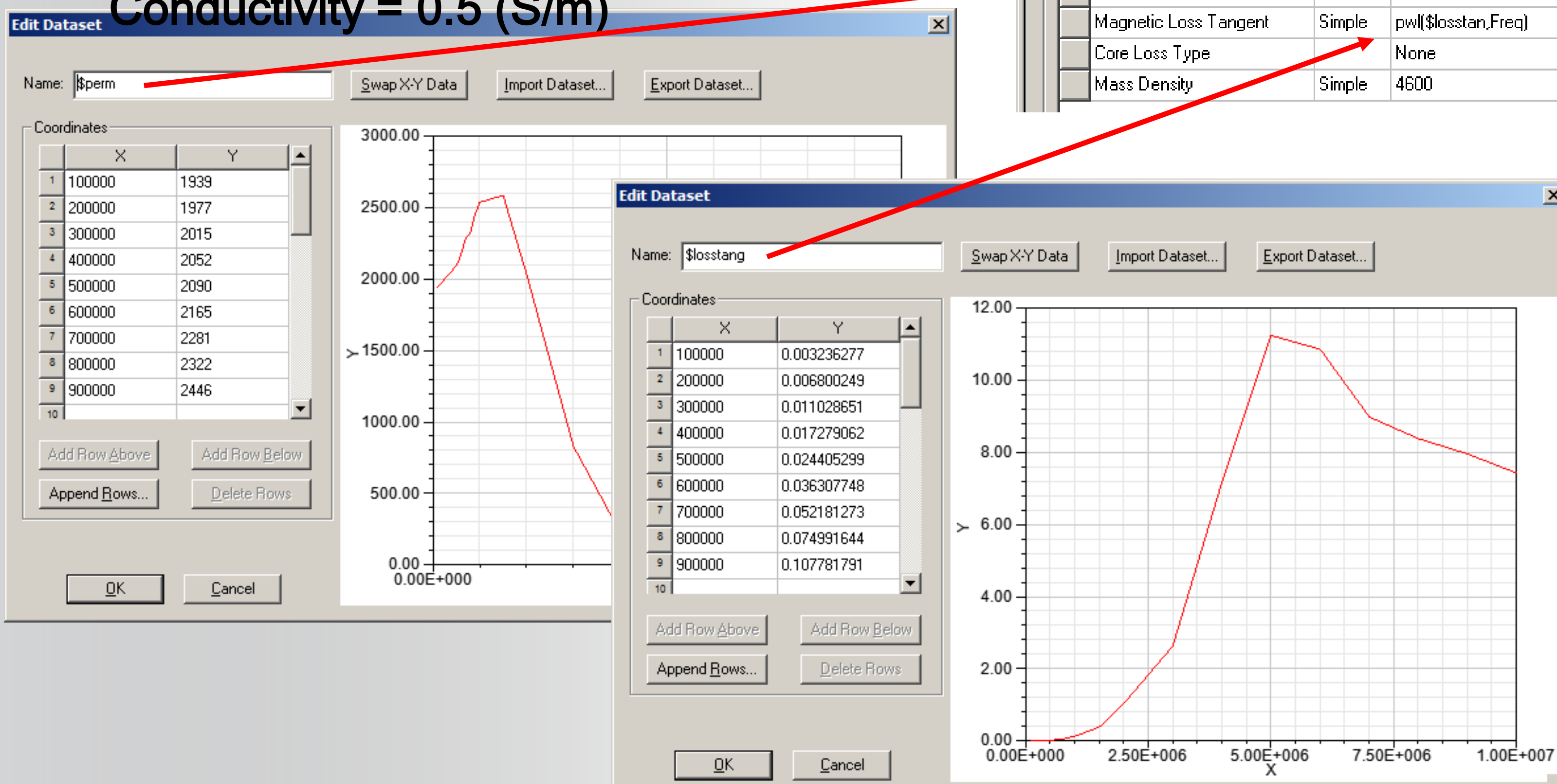
Datasets used to define properties vs. frequency:

- Relative Permeability = pwl($perm,Freq)
- Magnetic Loss tangent = pwl($losstan,Freq)

Relative permitivity = 12

Conductivity = 0.5 (S/m)



# 考虑温度特性的电导率材料属性

$$\sigma = \frac{1}{(1+0.0039*(Temp-22))}$$

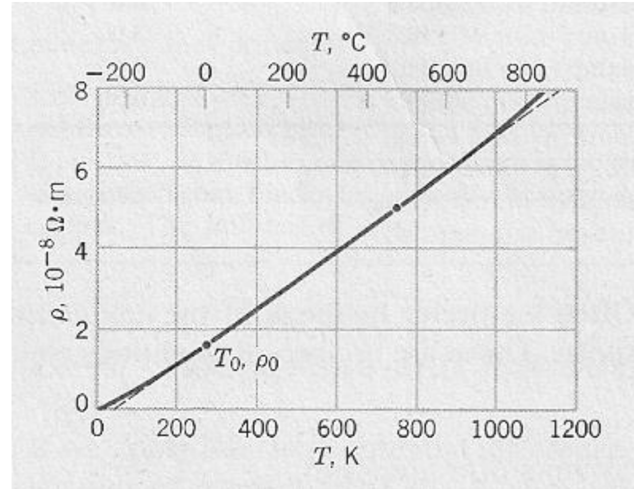
**View / Edit Material**

Material Name: copper_temp

Material Coordinate System Type: Cartesian

Properties of the Material

| Name | Type | Value | Units | Thermal Modifier |
| --- | --- | --- | --- | --- |
| Relative Permittivity | Simple | 1 | | None |
| Relative Permeability | Simple | 0.999991 | | None |
| Bulk Conductivity | Simple | 58000000 | siemens/m | 1 / (1 + 0.0039 * (Temp - 22)) |
| Dielectric Loss Tangent | Simple | 0 | | None |
| Magnetic Loss Tangent | Simple | 0 | | None |
| Core Loss Type | | None | w/m^3 | |
| Mass Density | Simple | 8933 | kg/m^3 | None |




**Edit Thermal Modifier**

Expression    Quadratic

Expression

Temperature-Dependent Bulk Conductivity:

P(Temp) = Pref [Modifier]

Reference Bulk Conductivity:

Pref = 58000000siemens/m

Parameters

Modifier: 1 / (1 + 0.0039 * (Temp - 22))

ANSYS

# 考虑温度特性的磁导率材料属性

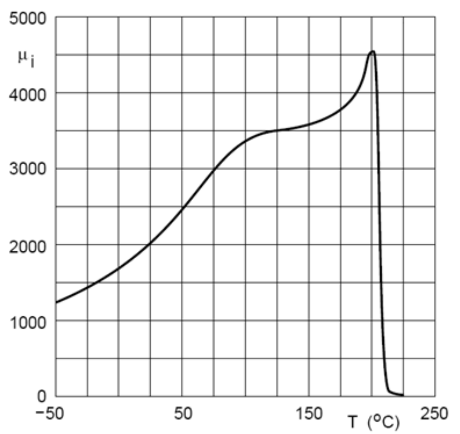
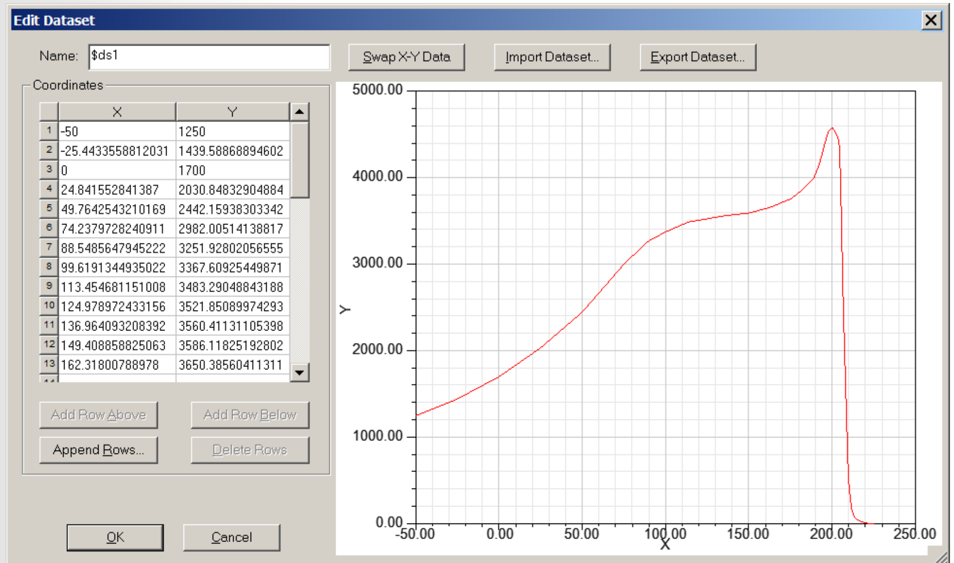
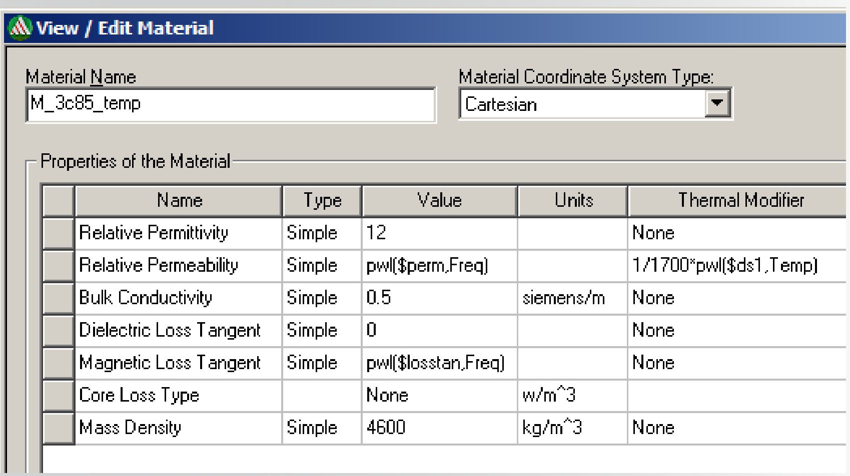
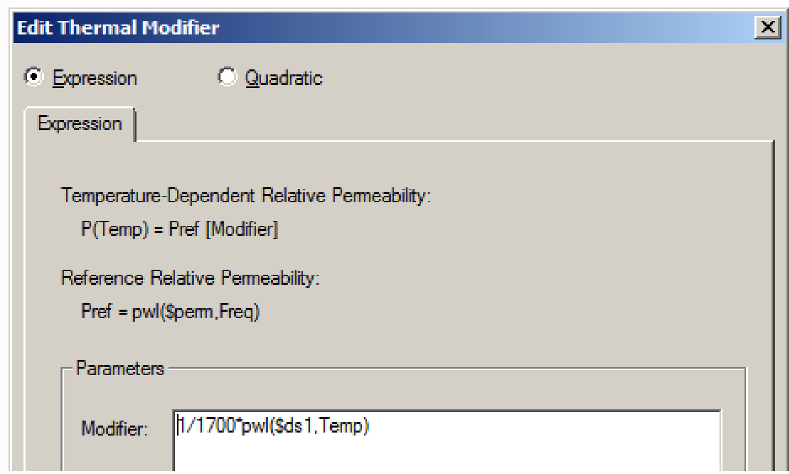



Fig.2 Initial permeability as a function of temperature.

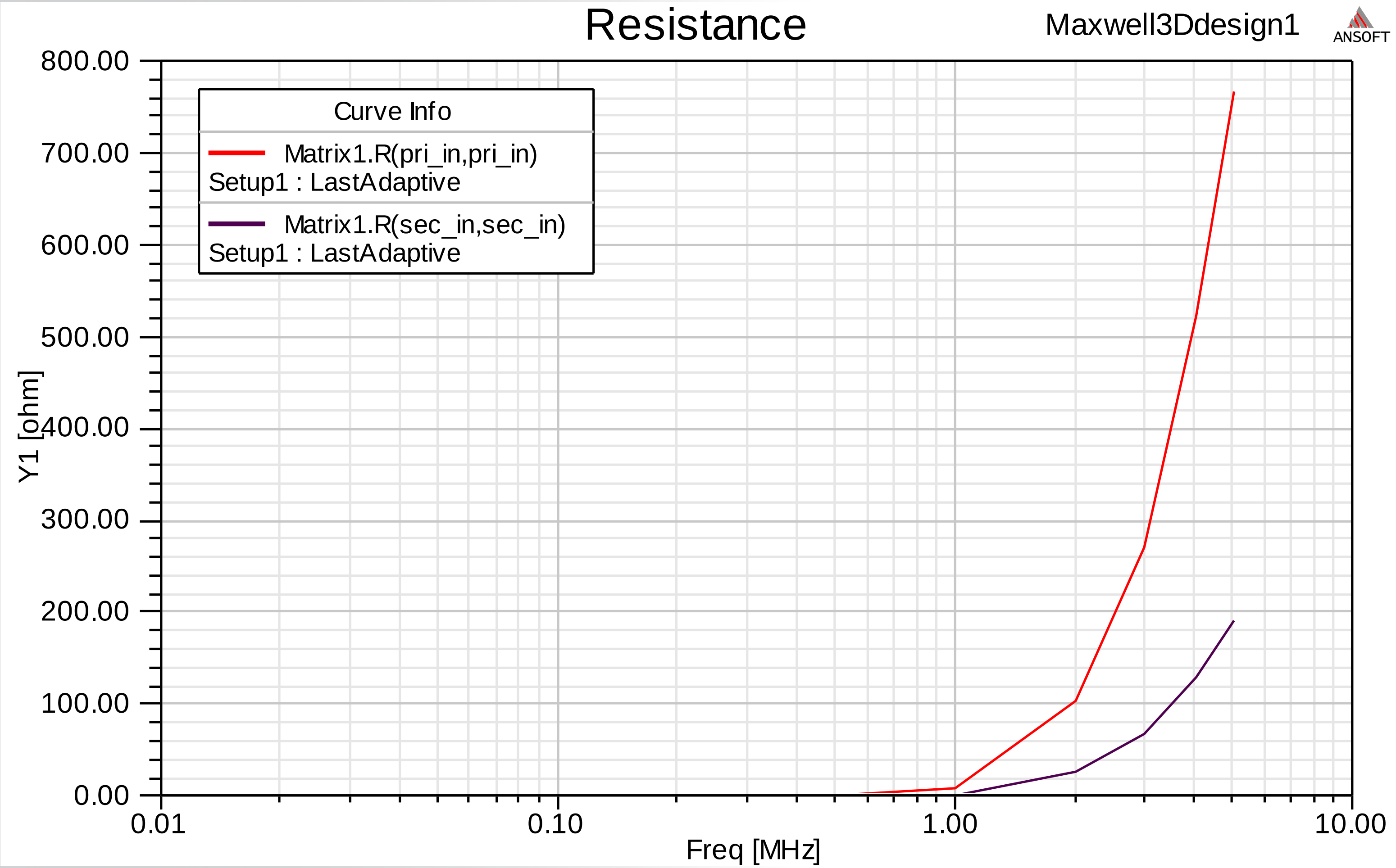


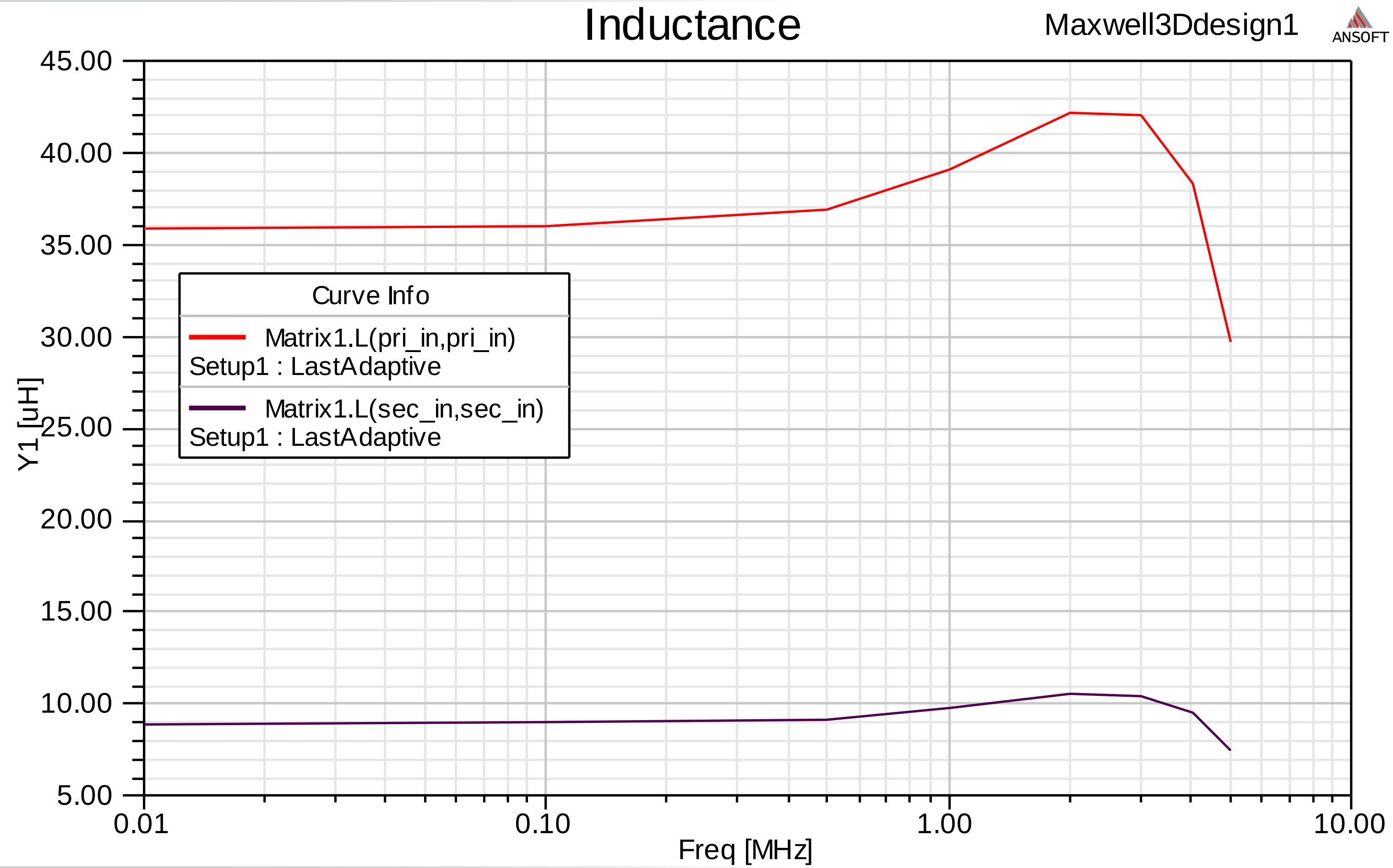
ANSYS

# 仿真结果


Resistance
Maxwell3Ddesign1
ANSOFT
Curve Info
Matrix1.R(pri_in,pri_in)
Setup1 : LastAdaptive
Matrix1.R(sec_in,sec_in)
Setup1 : LastAdaptive
Y1 [ohm]
800.00
700.00
600.00
500.00
400.00
300.00
200.00
100.00
0.00
0.01
0.10
1.00
10.00
Freq [MHz]

# 仿真结果

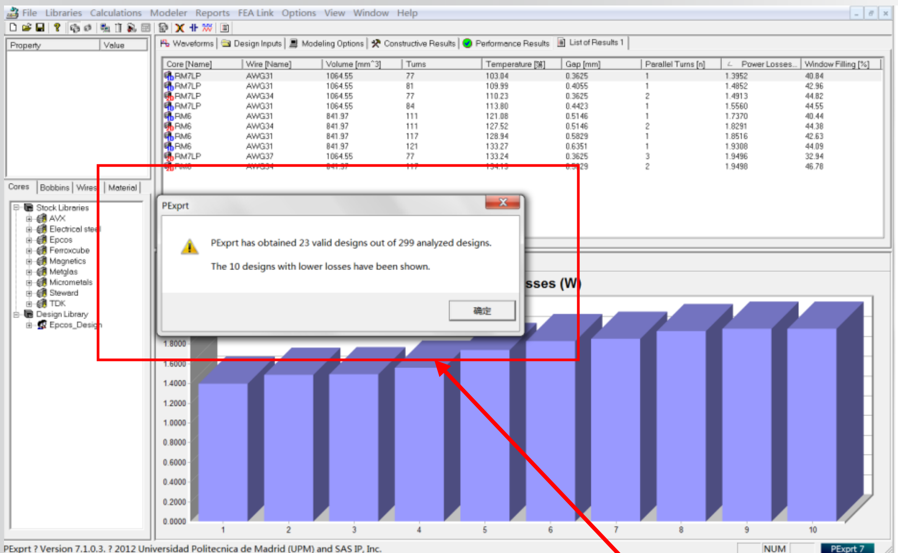

Inductance
Maxwell3Ddesign1
ANSOFT
45.00
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
Y1 [uH]
0.01
0.10
1.00
10.00
Freq [MHz]
Curve Info
Matrix1.L(pri_in,pri_in)
Setup1 : LastAdaptive
Matrix1.L(sec_in,sec_in)
Setup1 : LastAdaptive

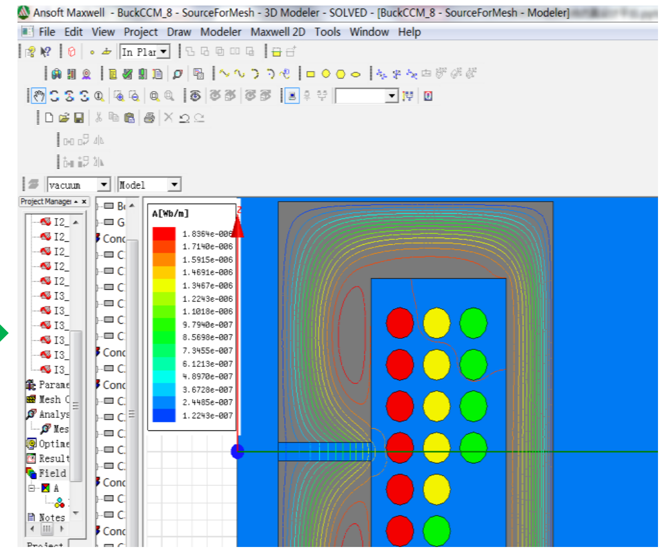
# 电磁部件设计建模（变压器，电感器）

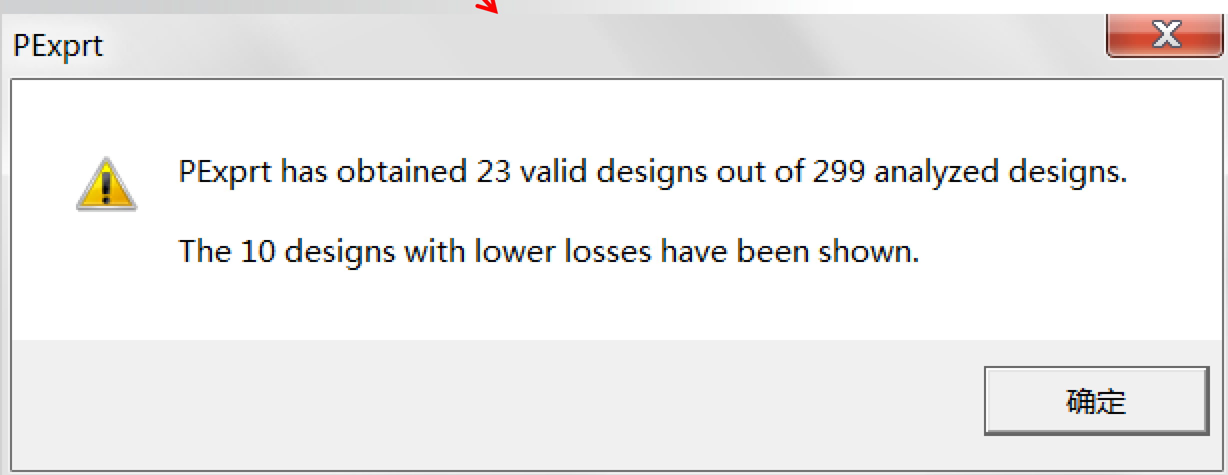
**PExprt、Maxwell 应用场合区别**

# 电磁部件设计建模（变压器，电感器）

## PExprt进行大范围搜素，Maxwell进行精确分析计算

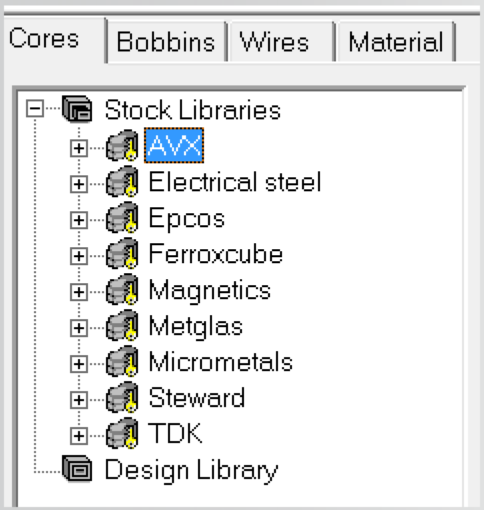


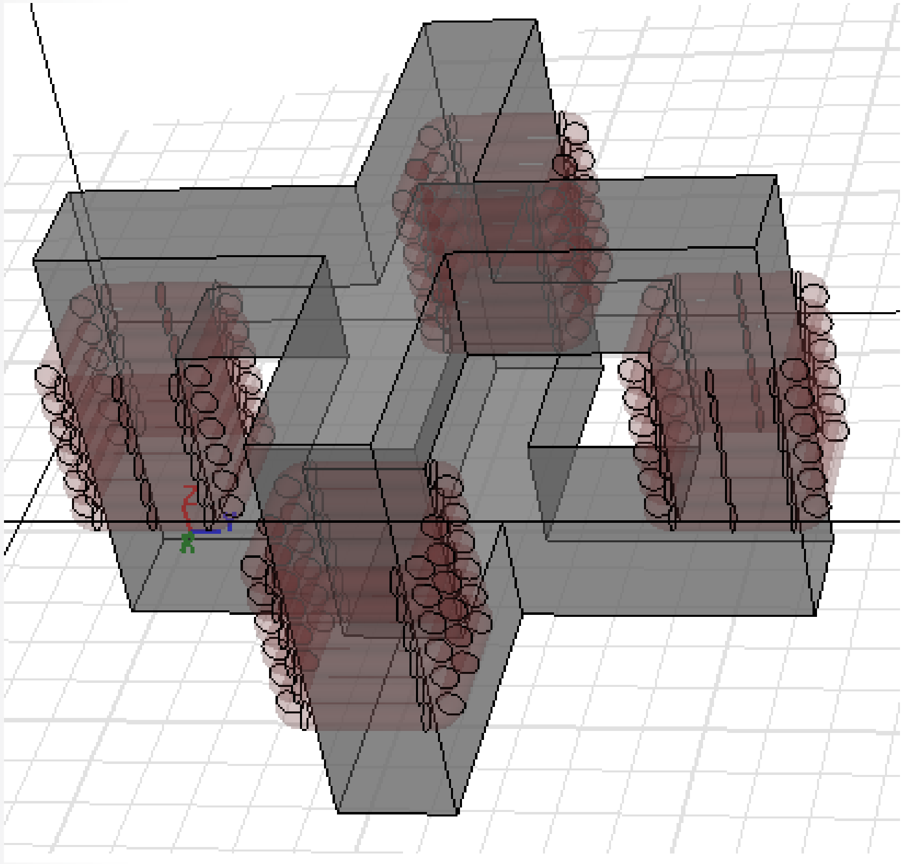



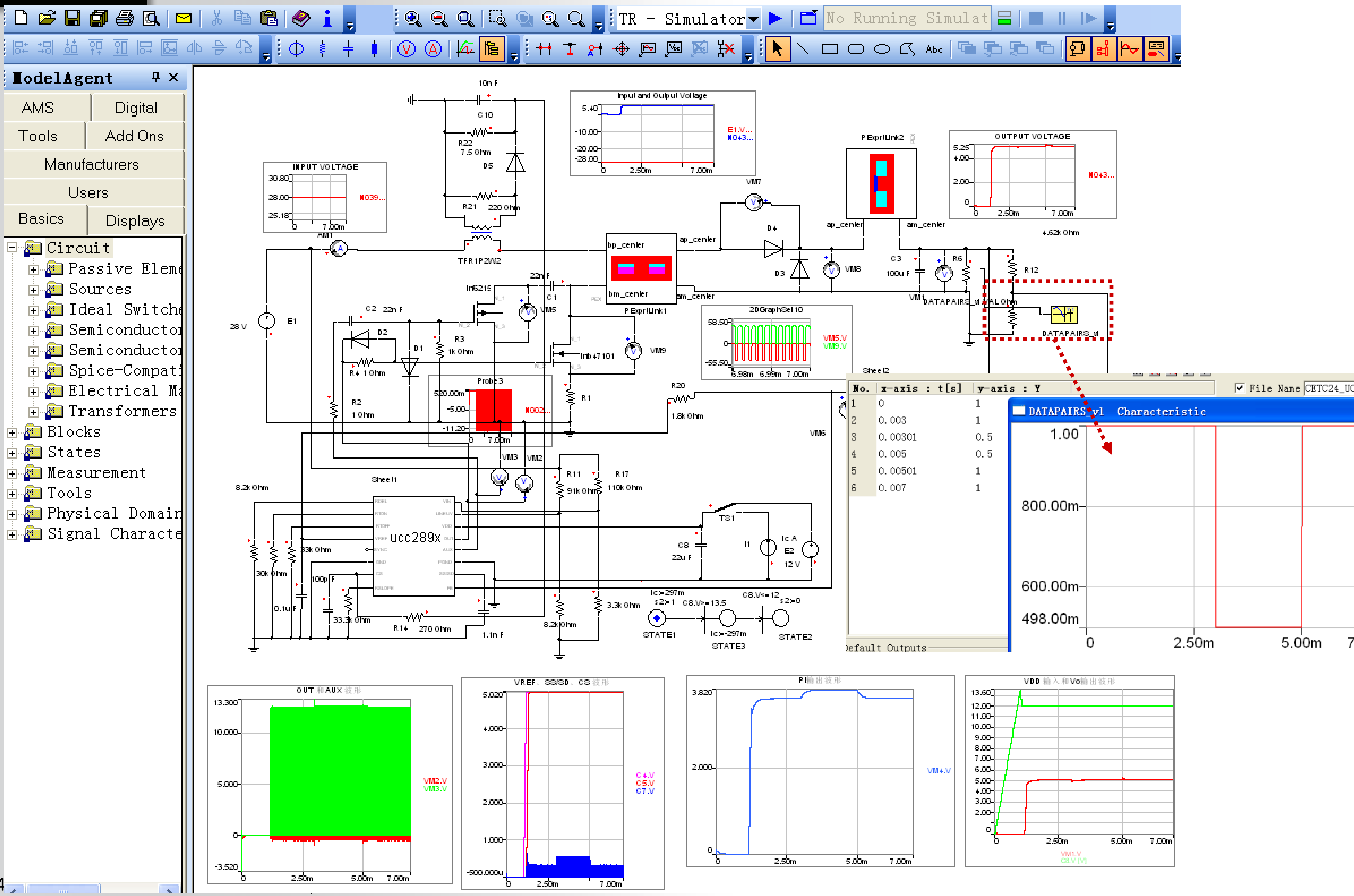
# 电磁部件设计建模（变压器， 电感器）

**PExprt设计形状规则电感、变压器**
**Maxwell设计任意电感、变压器**

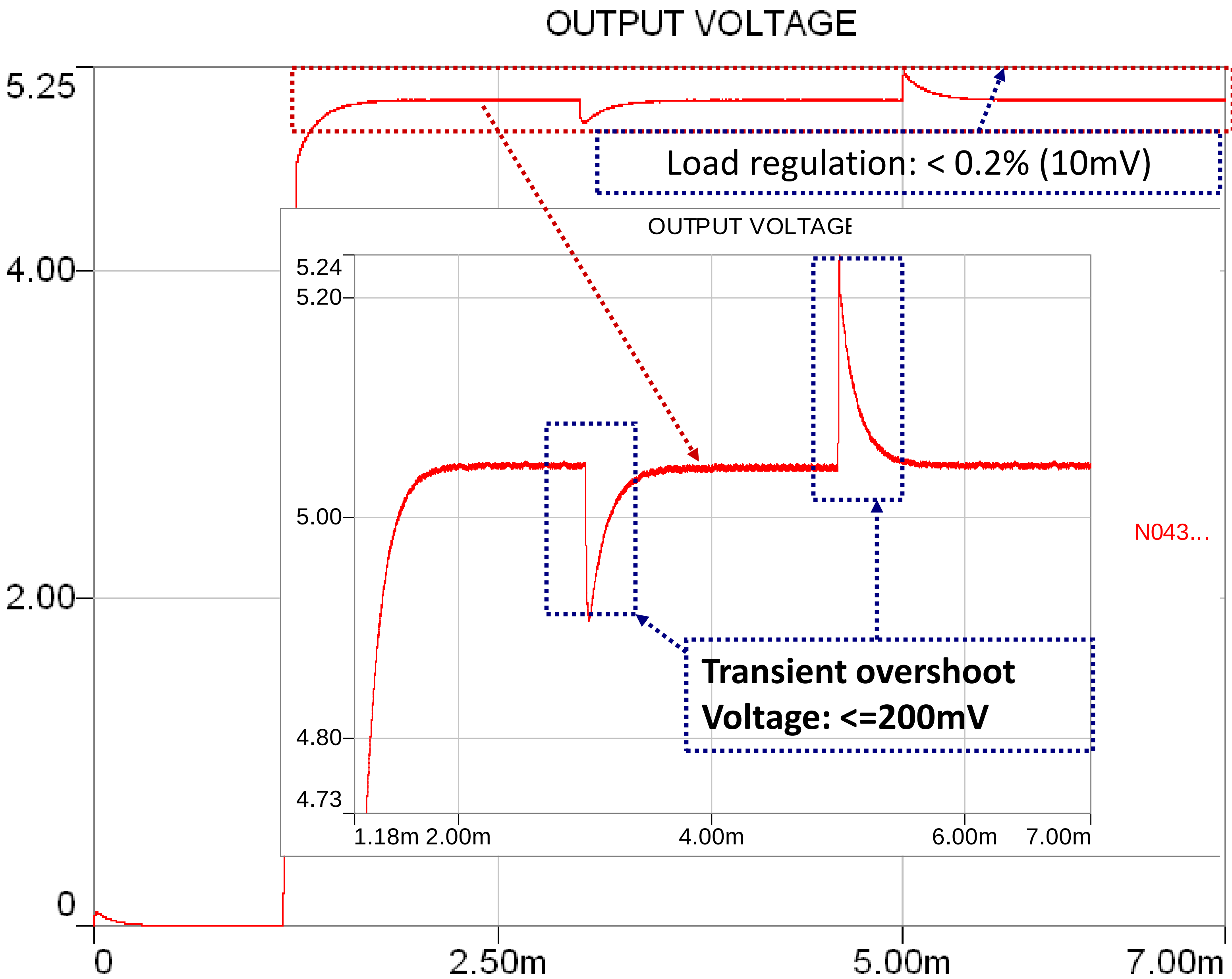


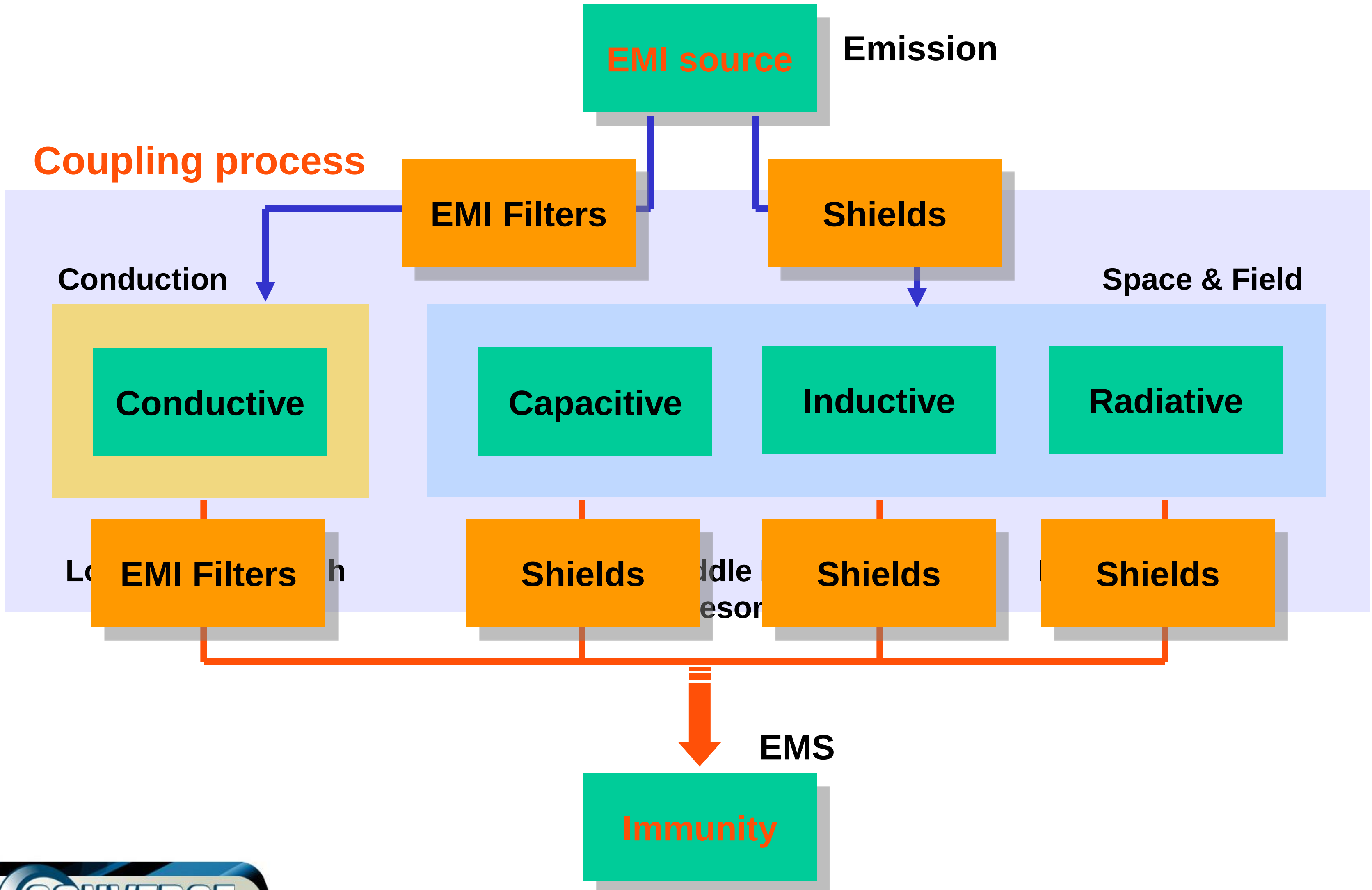
# 电磁部件与电路协同系统仿真



# 电磁部件与电路协同系统仿真

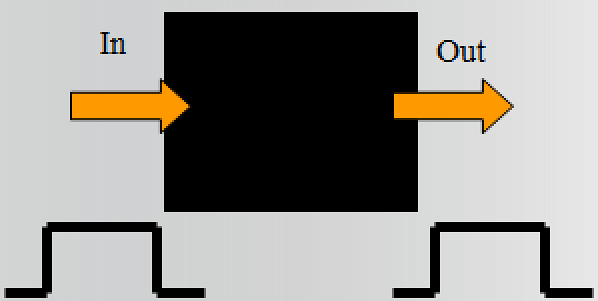


# 开关电源电磁兼容仿真


EMI source
Emission
Coupling process
EMI Filters
Shields
Conduction
Space & Field
Conductive
Capacitive
Inductive
Radiative
EMI Filters
Shields
Shields
Shields
EMS
Immunity

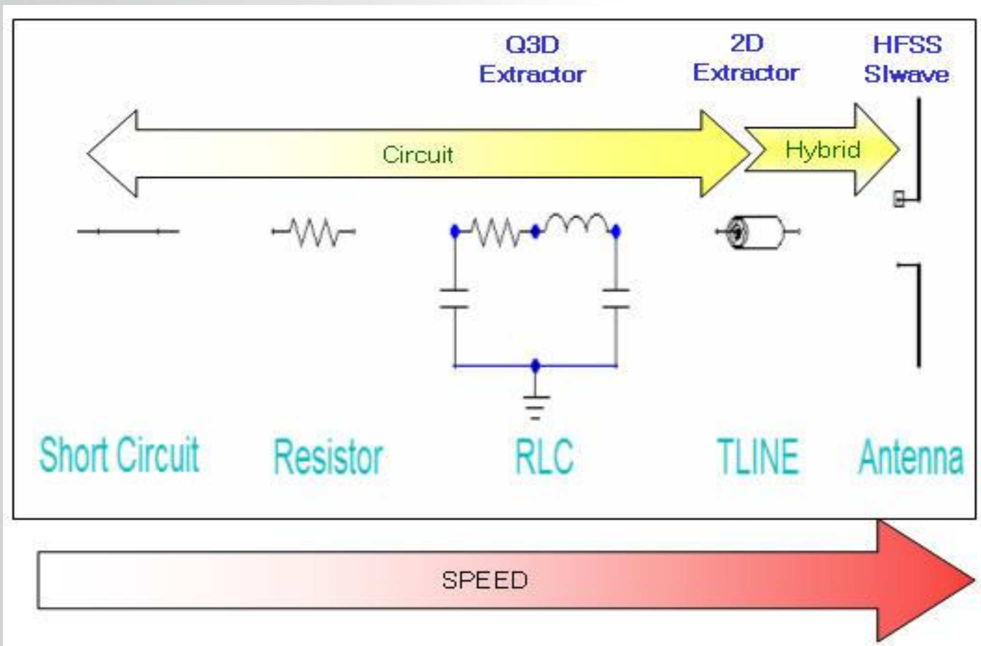
# 仿真解决电磁干扰关键点

**模型建立：建模体现源头、耦合路径、受扰体的细节**

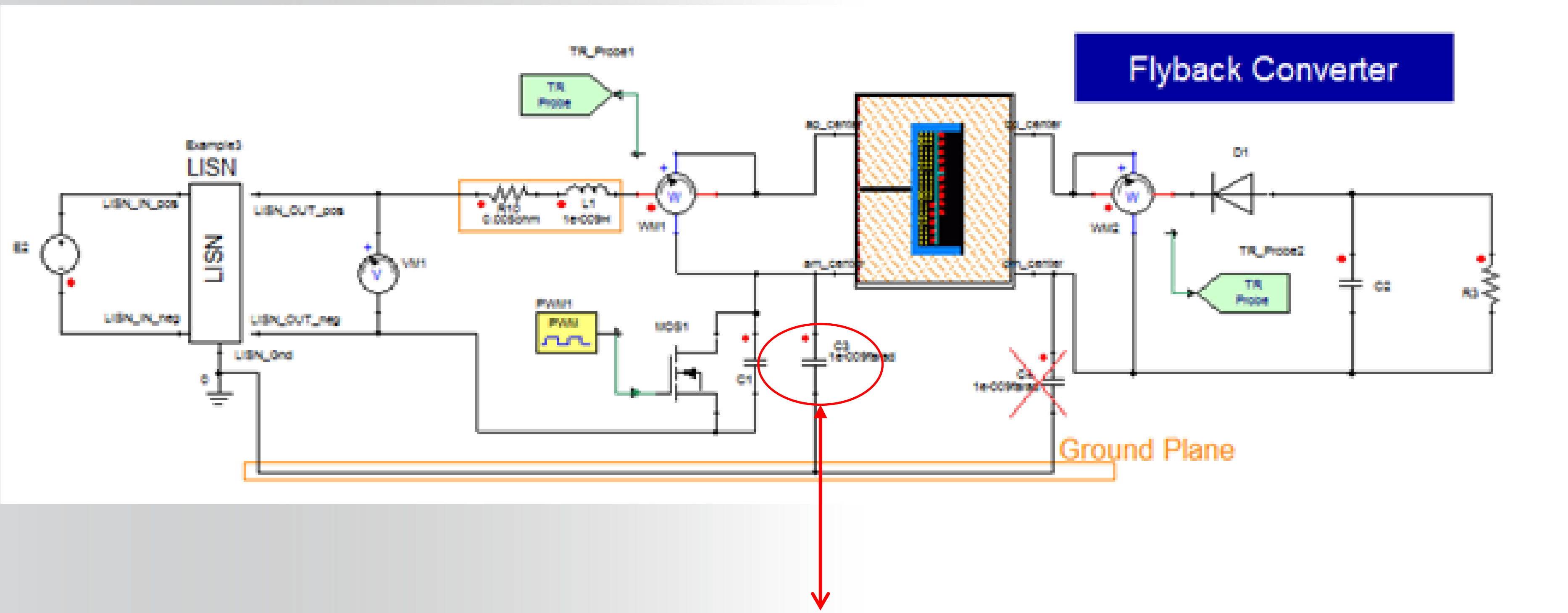



理想电气线路


考虑寄生参数的电气



# 仿真解决电磁干扰关键点

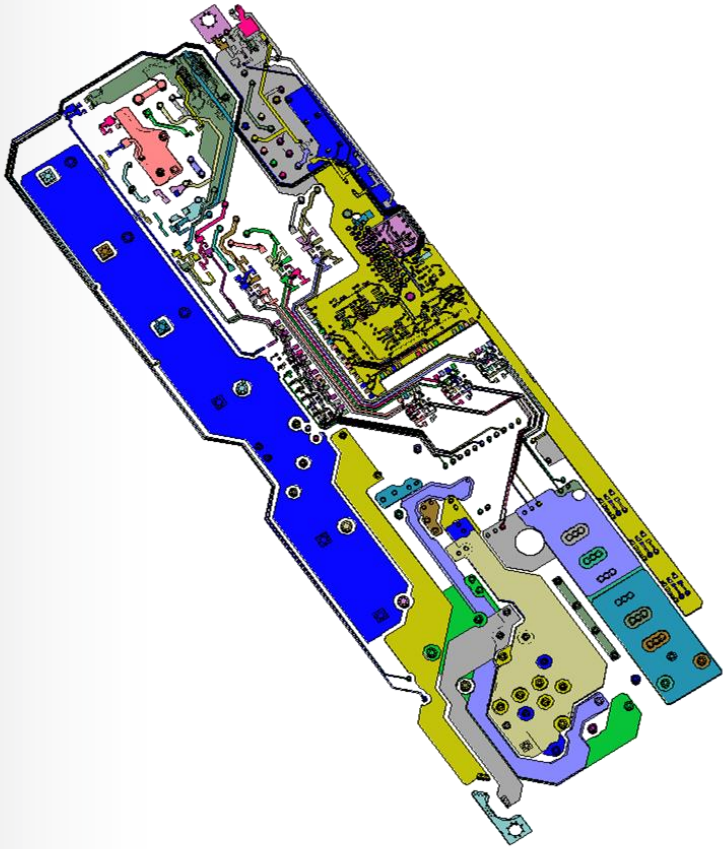



对地电容是共模电流的主要通道，仿真必须能体现该电容

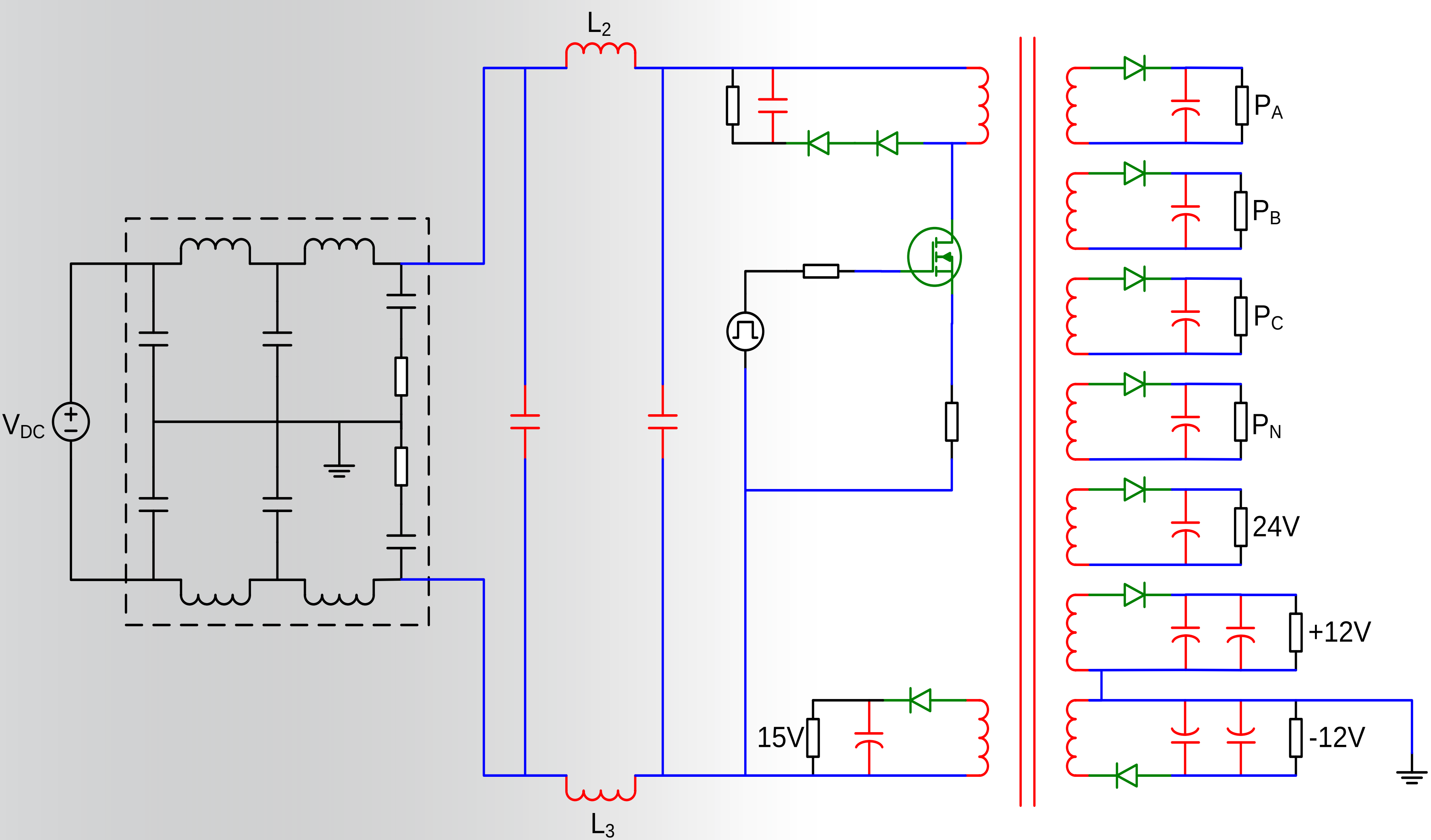
# Q3D Extractor – 简介

- Q3D **Extractor** 是电路板、连接器、母排和线缆电气寄生参数的快速提取工具.
- 与常用的布板工具有接口
  - Cadence
  - Mentor
- Q3D 包含两个工具:
  - Q3D Extractor
  - 2D Extractor

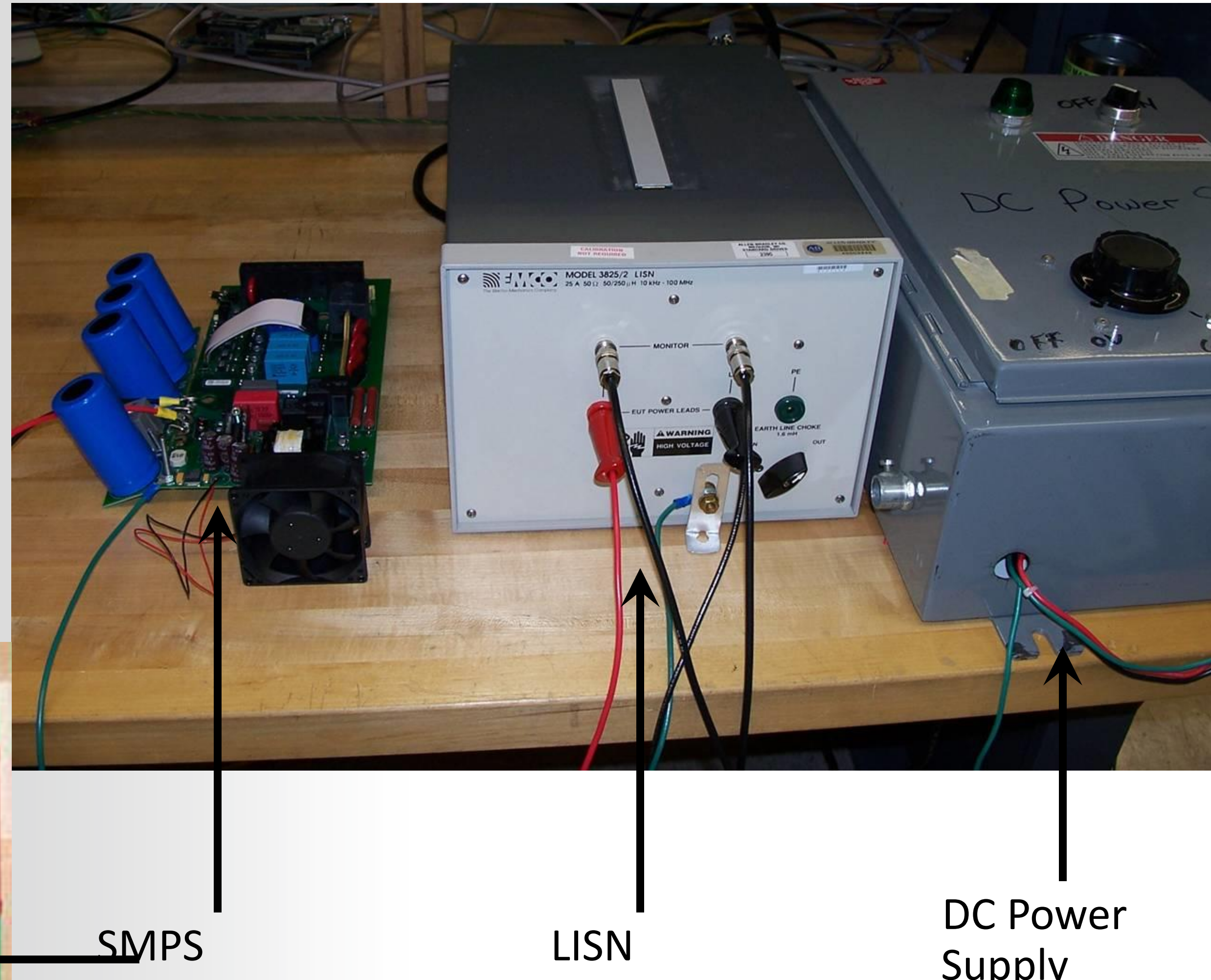
# 开关电源电磁兼容仿真



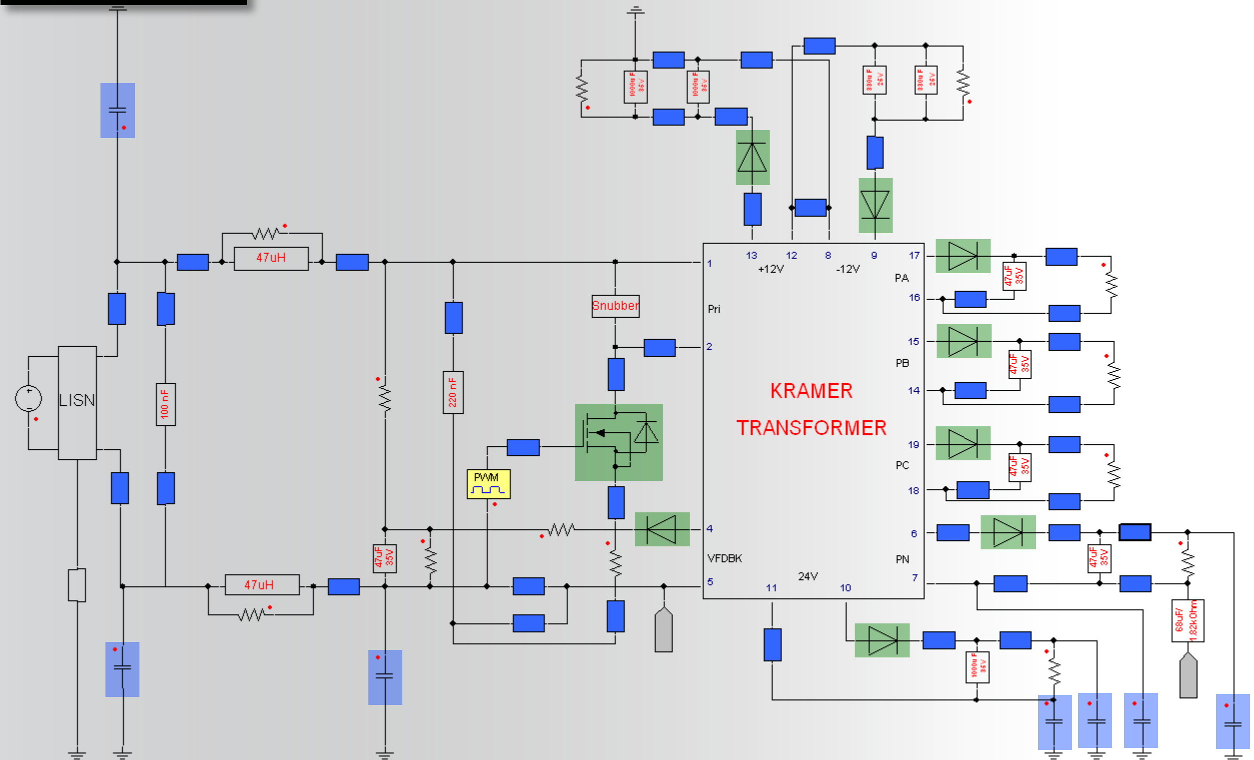

  

# 开关电源电磁兼容仿真

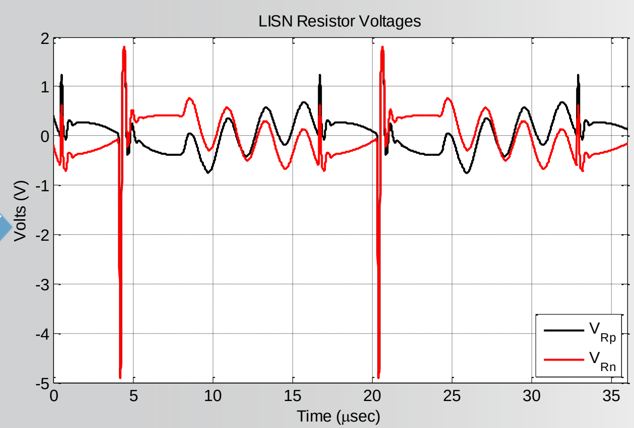
- 125 MHz sampling
- 10 ms recorded
- $V_{DC}$=200V nominally (up to 600V)
- 50% (DC fan) load nominally

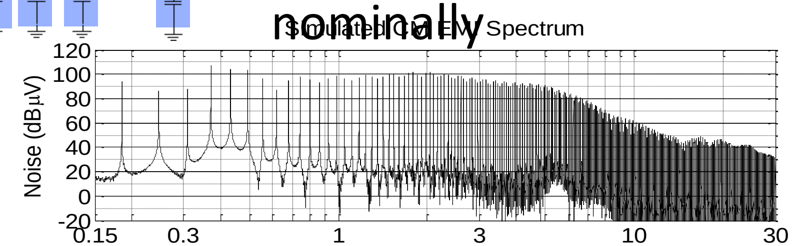


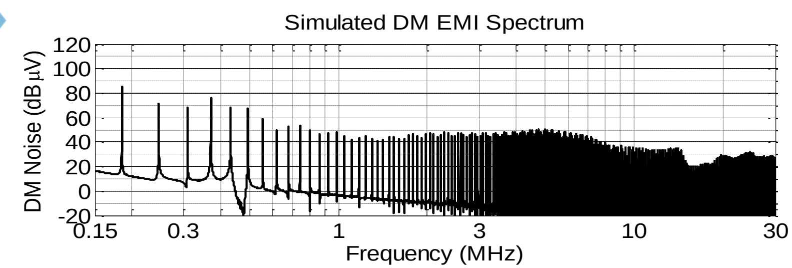

# 开关电源电磁兼容仿真



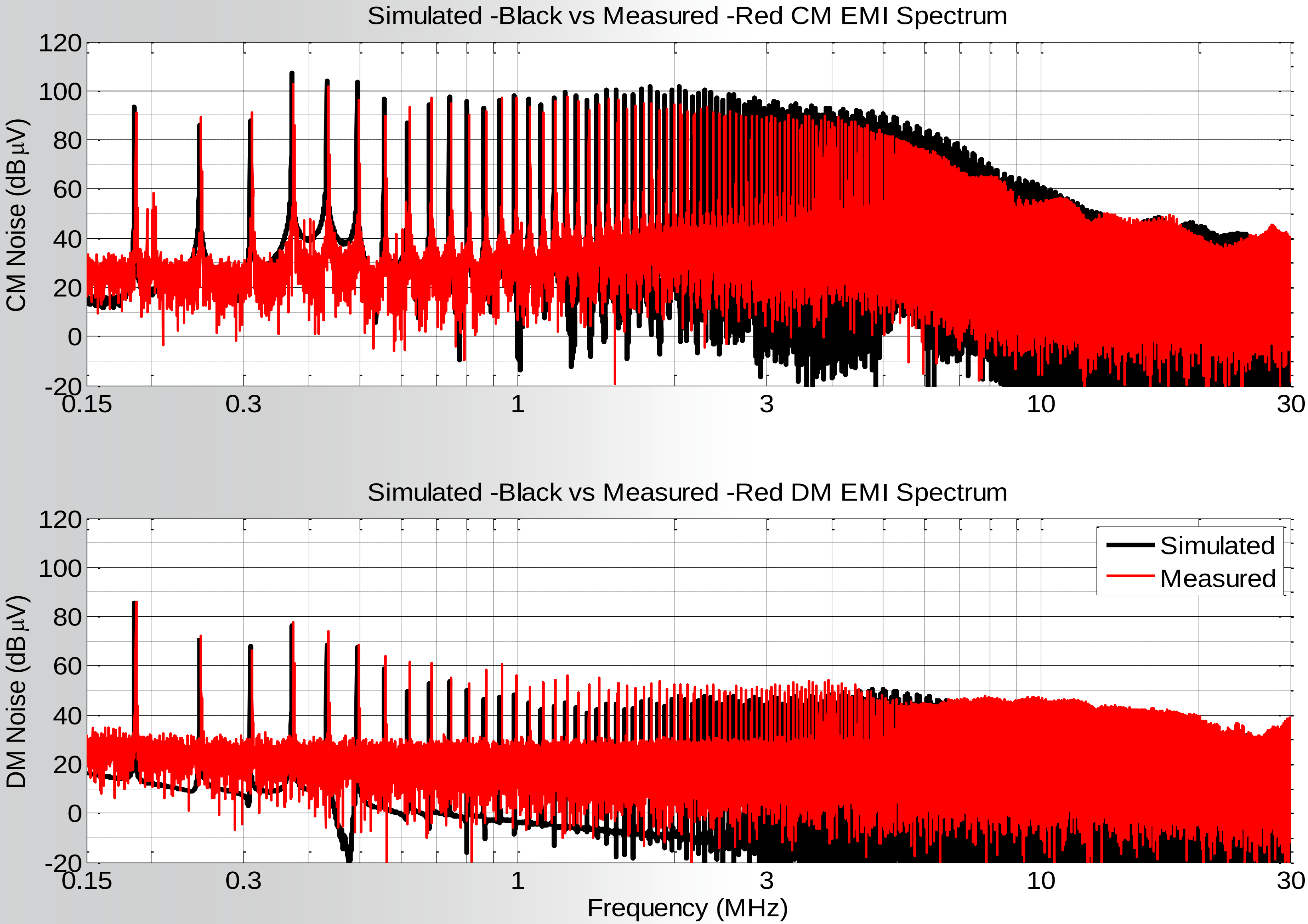

- Blue elements are manually input from Q3D and are turned on/off
- 2 ns step size
- 4 ms total time
- 30-60 min Simulation time
- $V_{DC}$=200V nominally
- 50% (resistive) load nominally



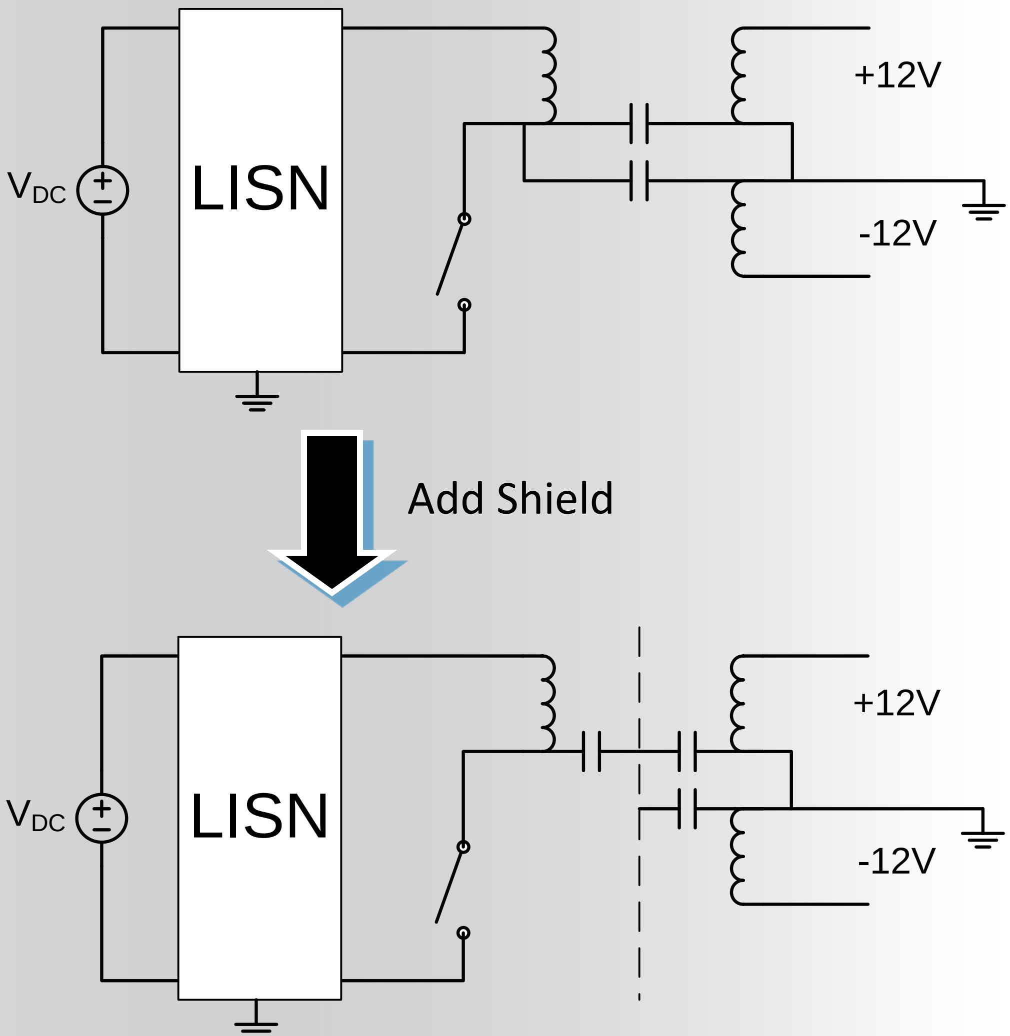



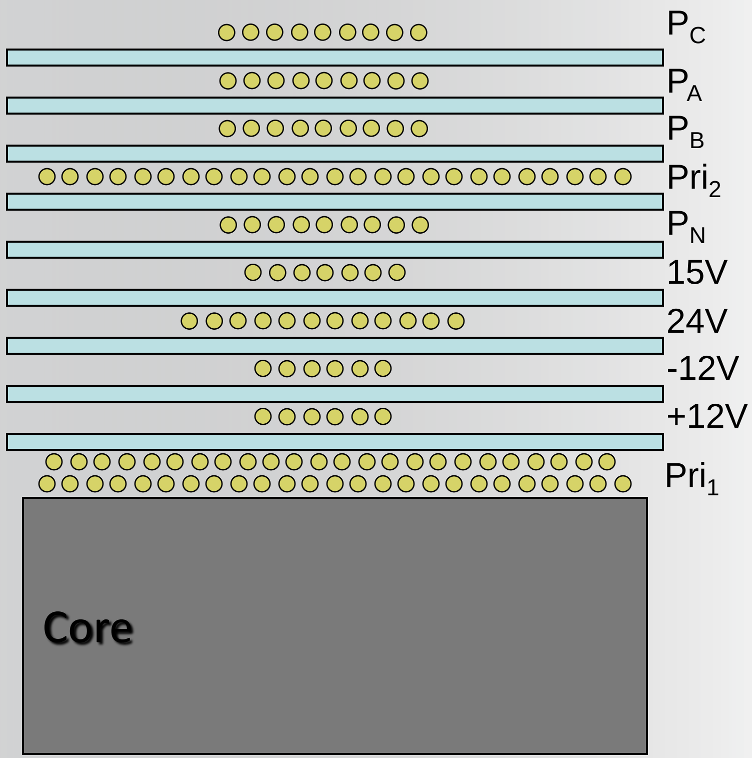
# 开关电源电磁兼容仿真


Simulated -Black vs Measured -Red CM EMI Spectrum
CM Noise (dBµV)
120
100
80
60
40
20
0
-20
0.15
0.3
1
3
10
30
Simulated -Black vs Measured -Red DM EMI Spectrum
DM Noise (dBµV)
Simulated
Measured
Frequency (MHz)

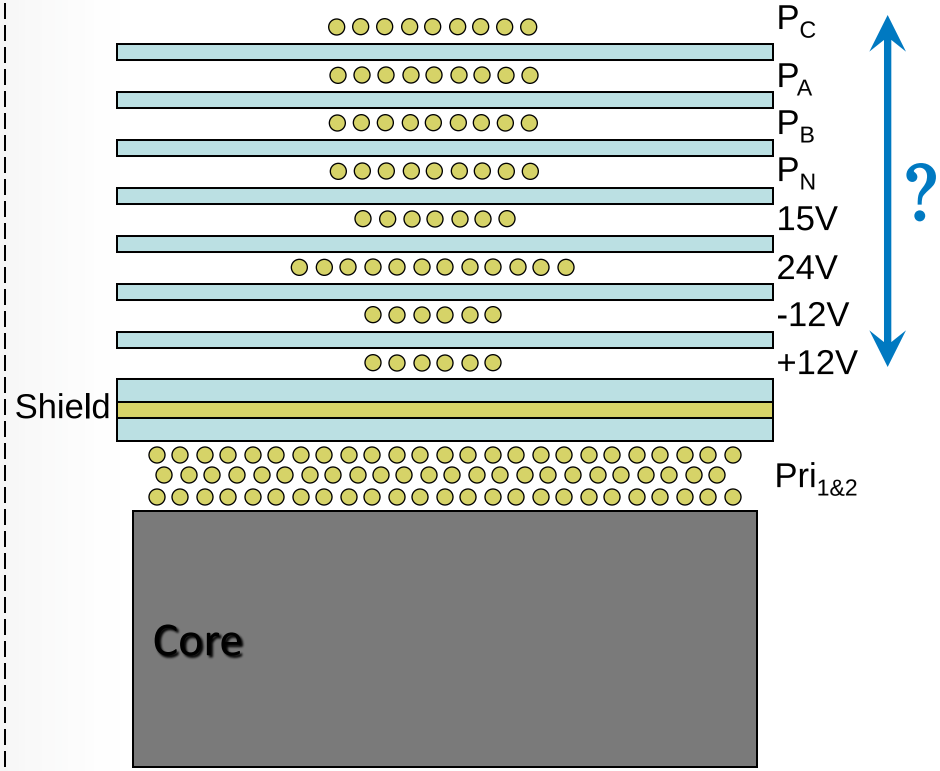
  



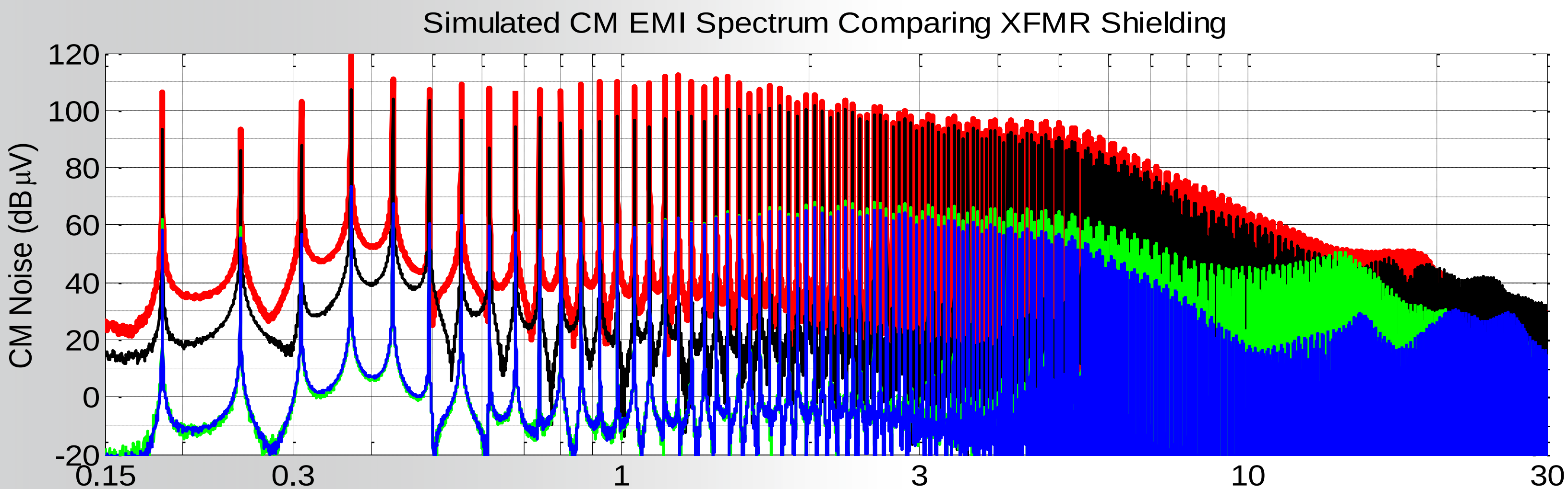

- Where/how to physically place shield
- Where to electrically connect shield
- Size and cost of shield
- Is it all worth it?



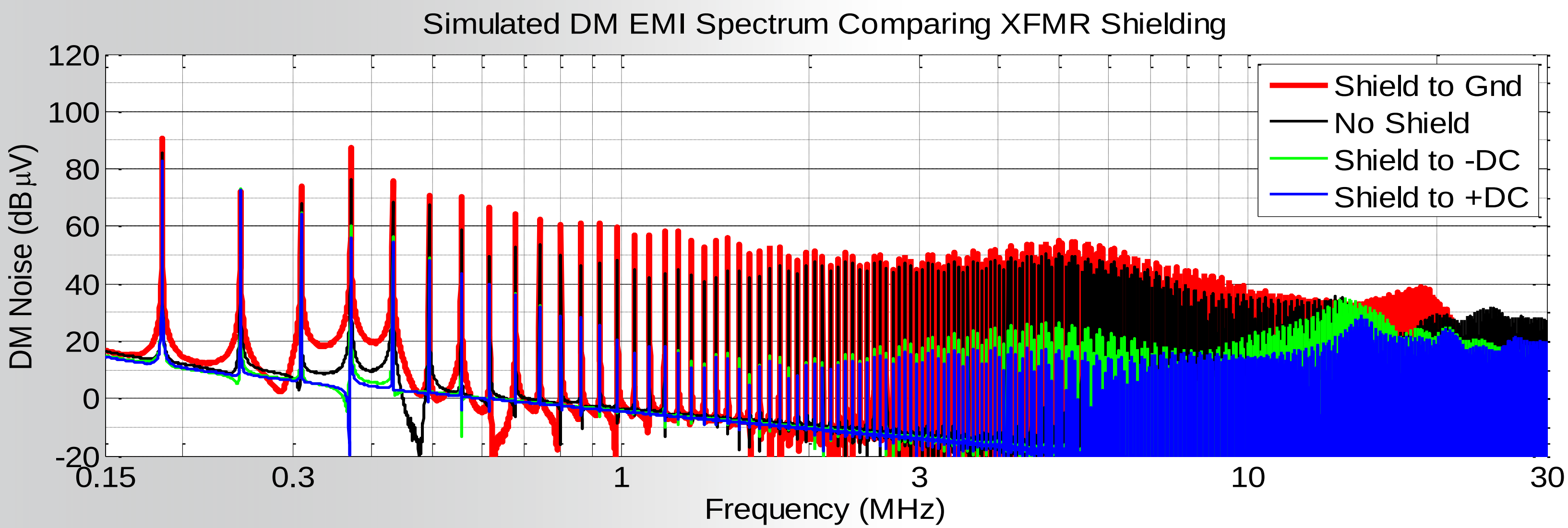

Present
Winding Cross-Section




Proposed (Shielded)
Winding Cross-Section



Simulated CM EMI Spectrum Comparing XFMR Shielding
CM Noise (dB µV)
120
100
80
60
40
20
0
-20
0.15
0.3
1
3
10
30

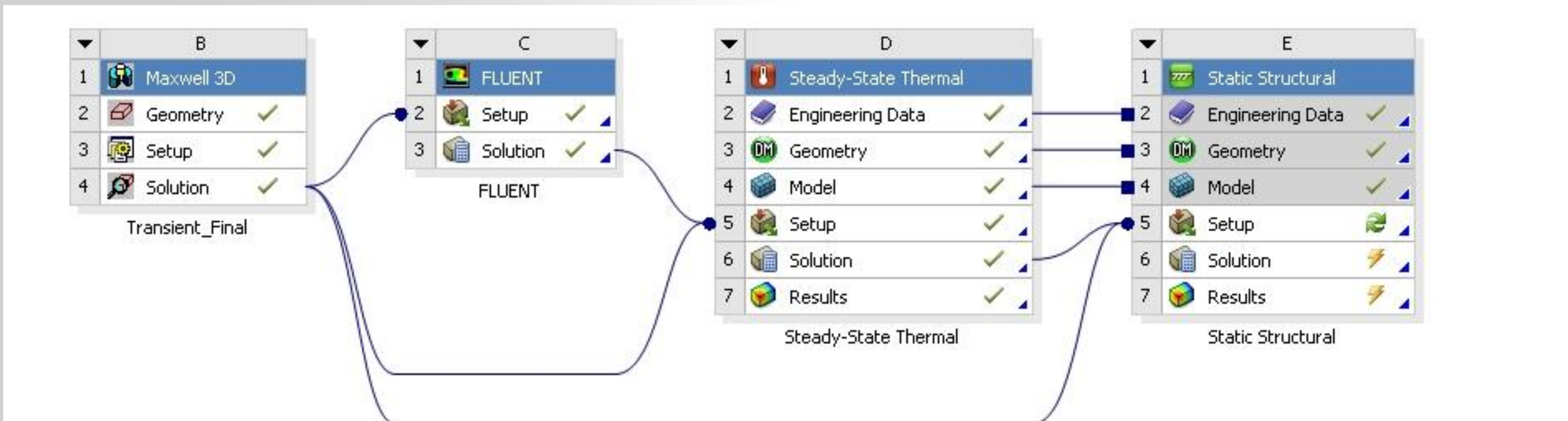



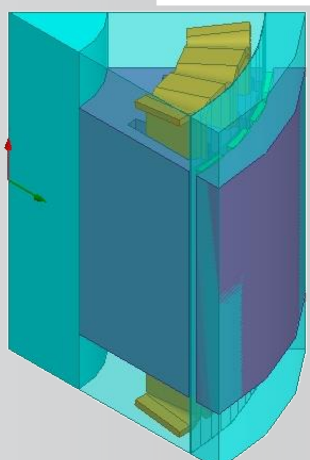
Simulated DM EMI Spectrum Comparing XFMR Shielding
DM Noise (dB µV)
120
100
80
60
40
20
0
-20
0.15
0.3
1
3
10
30
Frequency (MHz)
Shield to Gnd
No Shield
Shield to -DC
Shield to +DC

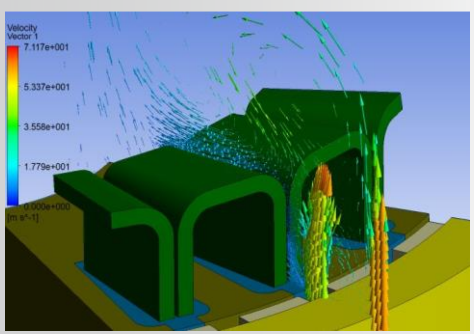

  

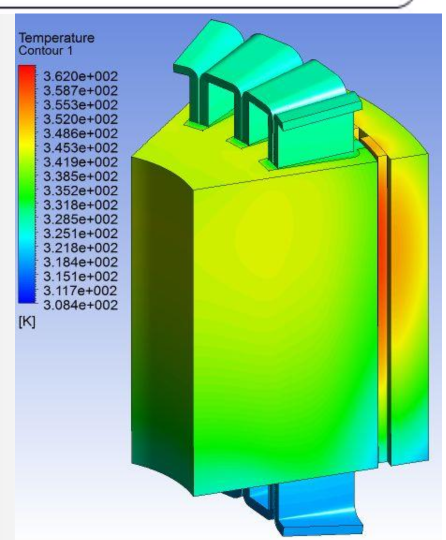
# 电源系统多物理设计

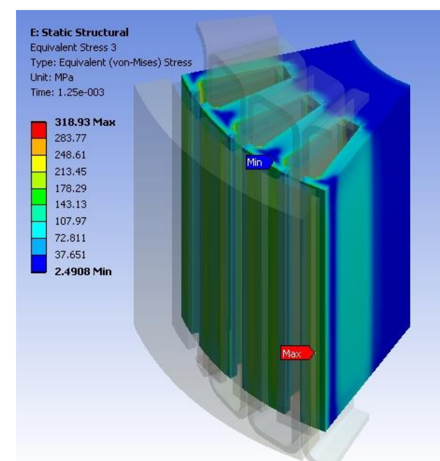
**ANSYS把各物理域软件集成到同一个平台Workbench下，各模块之间无缝实现数据共享和传输，相互之间还能迭代，使仿真模型最大限度接近物理实际模型**



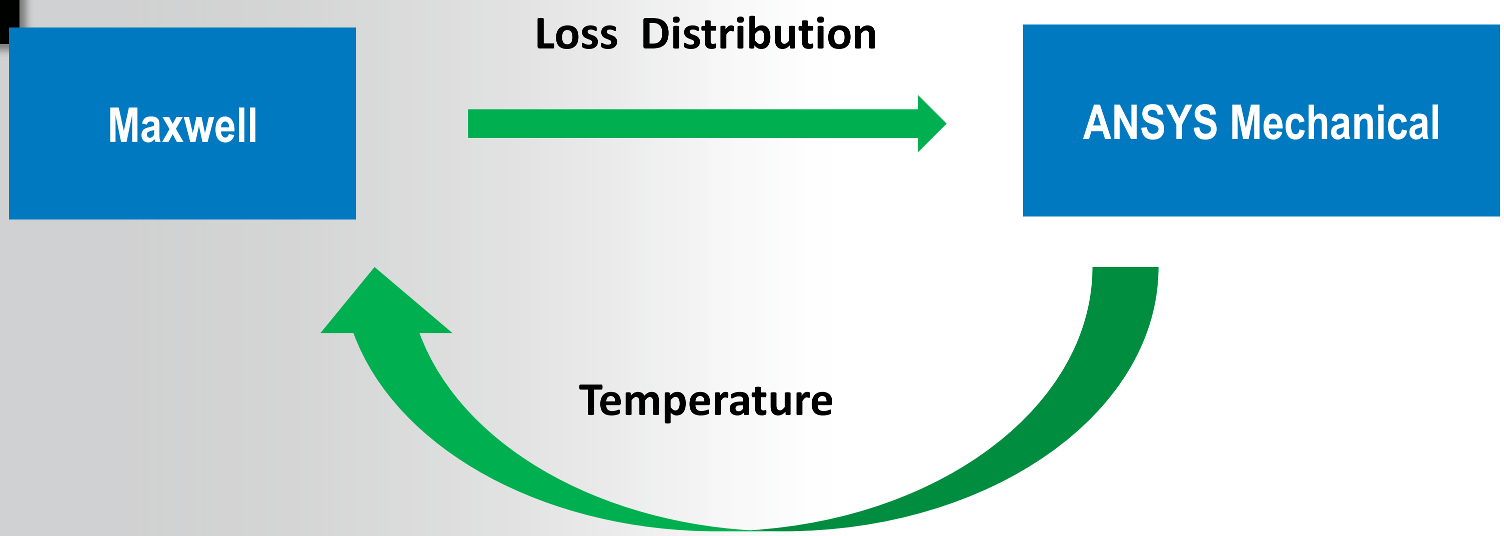





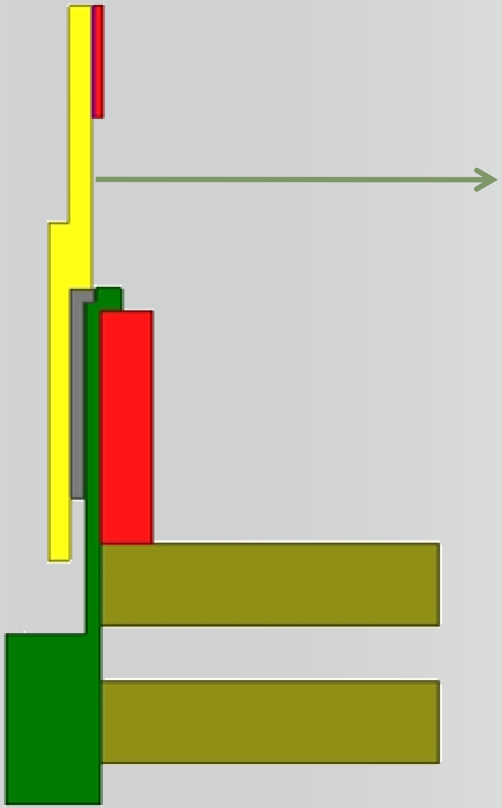



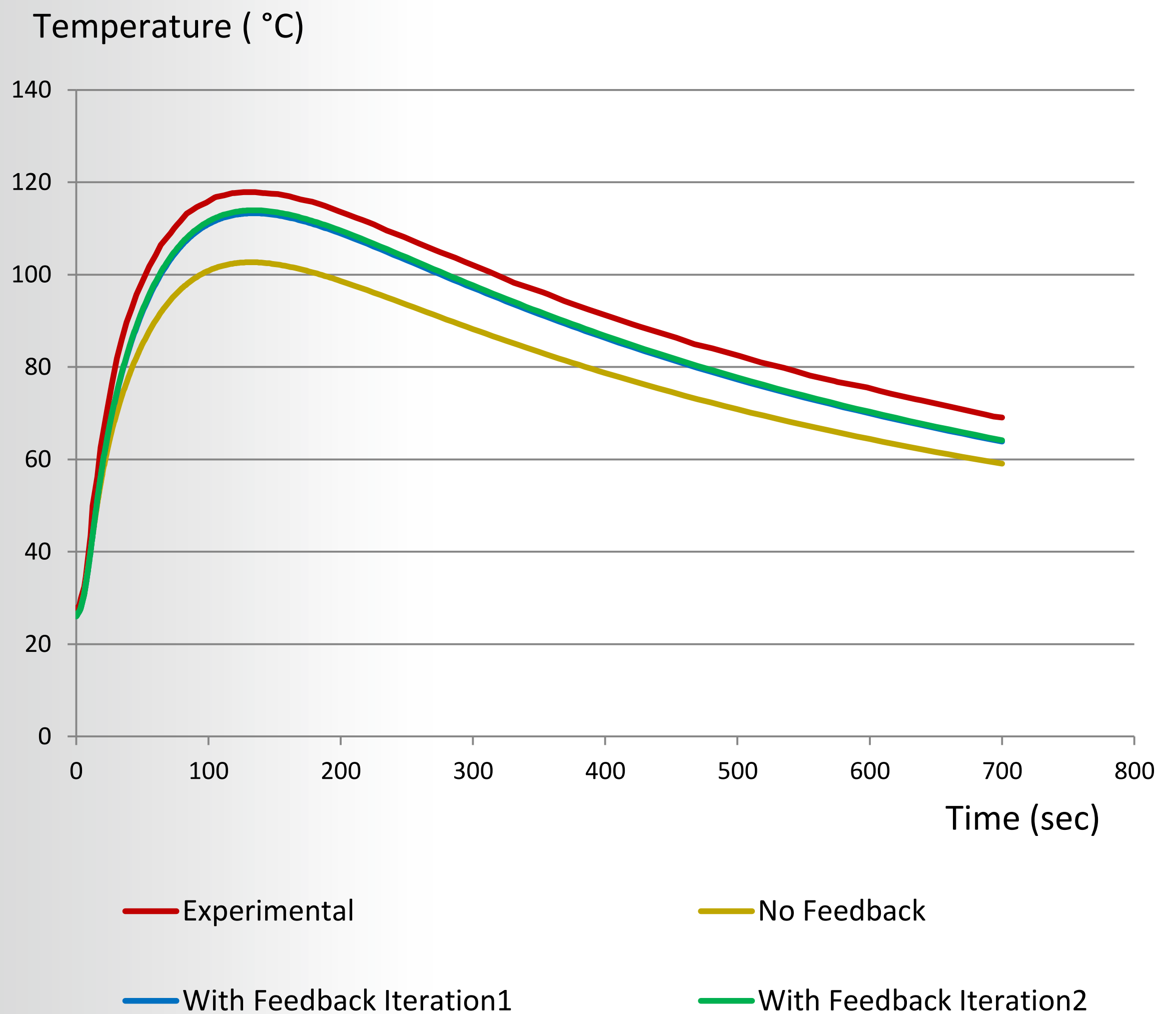
# Maxwell – ANSYS 温度场耦合




- Maxwell 中定义带有温度特性的原材料，设置初始温度，求解得到损耗分布，损耗分布无缝输入到ANSYS Mechanical中；
- ANSYS Mechanical根据损耗分布 求得温度分布，返回Maxwell中；
- Maxwell根据新得到的温度，改变材料特性，求得新的损耗，再与ANSYS Mechanical数据迭代，直到温度稳定

# 电源系统多物理设计




Temperature ( °C)
140
120
100
80
60
40
20
0
0
100
200
300
400
500
600
700
800
Time (sec)
Experimental
No Feedback
With Feedback Iteration1
With Feedback Iteration2

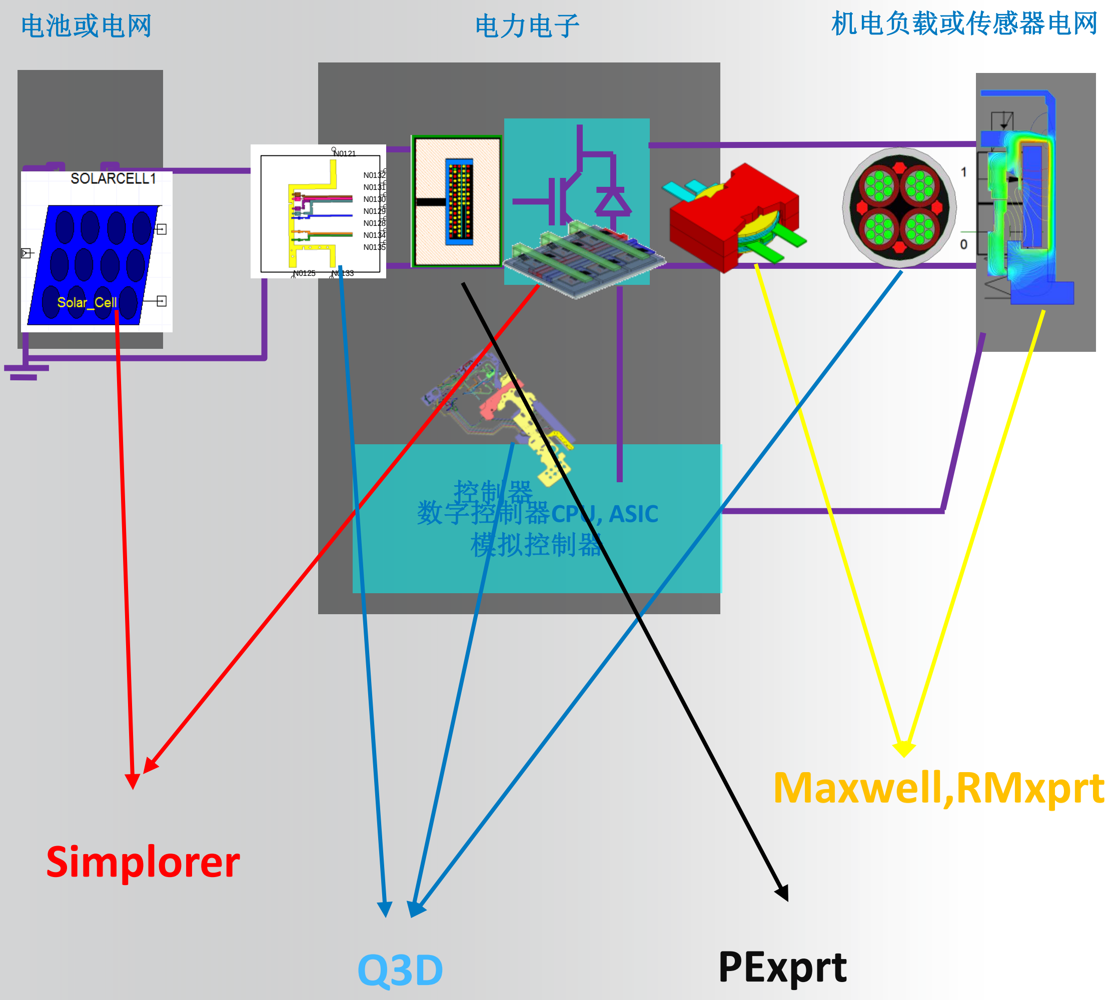
# 内容提要

ANSYS电源系统仿真设计理念

ANSYS电源系统仿真设计实现

总结

# ANSYS提供完整电源系统仿真设计平台方案




ANSYS 其它物理域软件 – 如ICEPAK、MECHANICAL、FLUENT

多种仿真工具协助工程师设计和分析开关器件、磁性器件、驱动系统、控制算法等、系统响应等；

可以进行开关电源电磁兼容仿真；

具备热、应力等多物理域仿真能力；

模块之间无缝接口，客户可以根据需求灵活配置；

软件平台包含成熟先进的电源设计思路和模型，相当于一个专家系统；

虚拟实验平台低成本，高效率，无限拓展测试项目和优化分析空间

让我们的技术助你们实现技术更新、智领先机！

协助用户取得附加价值才是我们最大的价值！